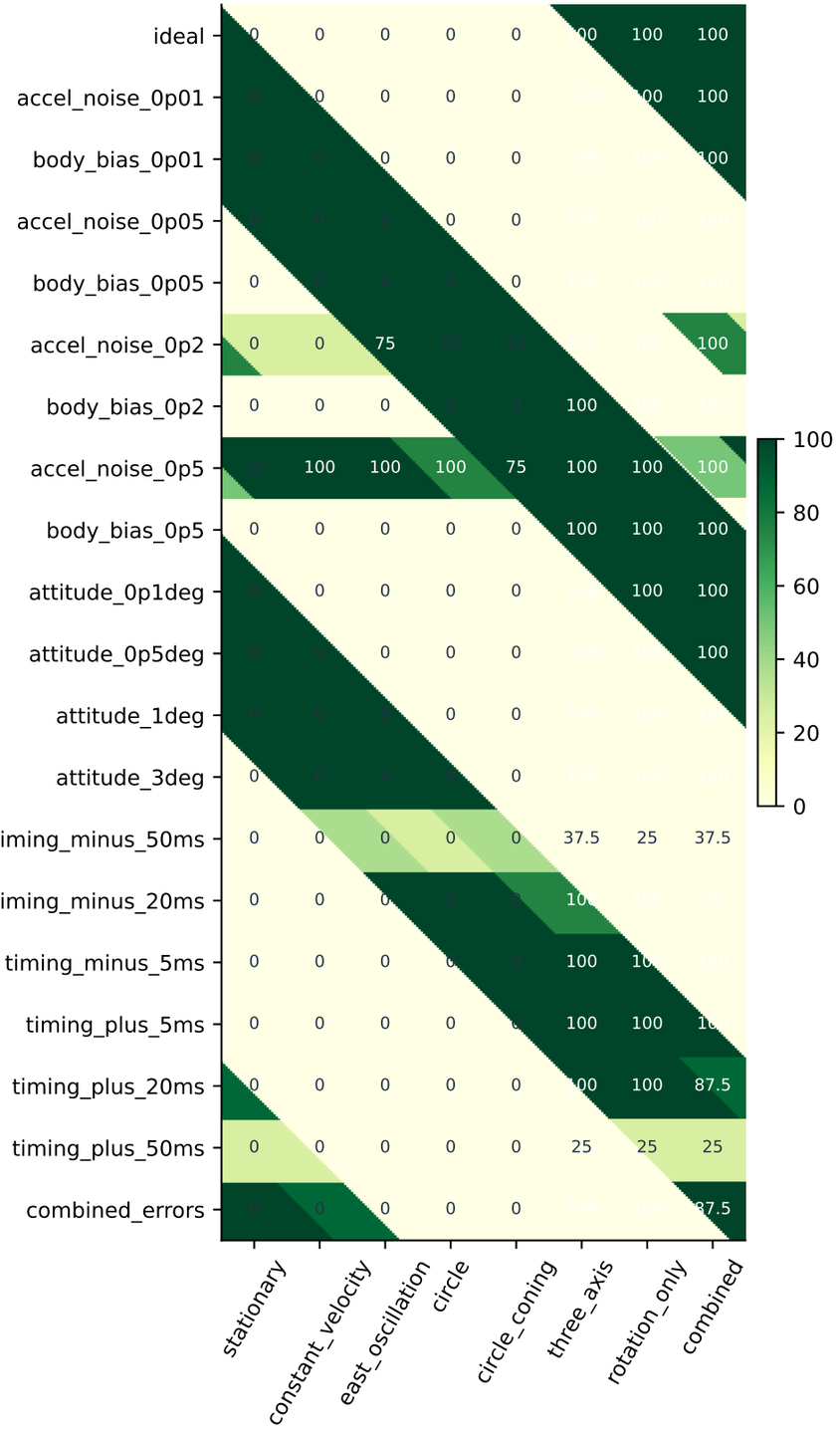
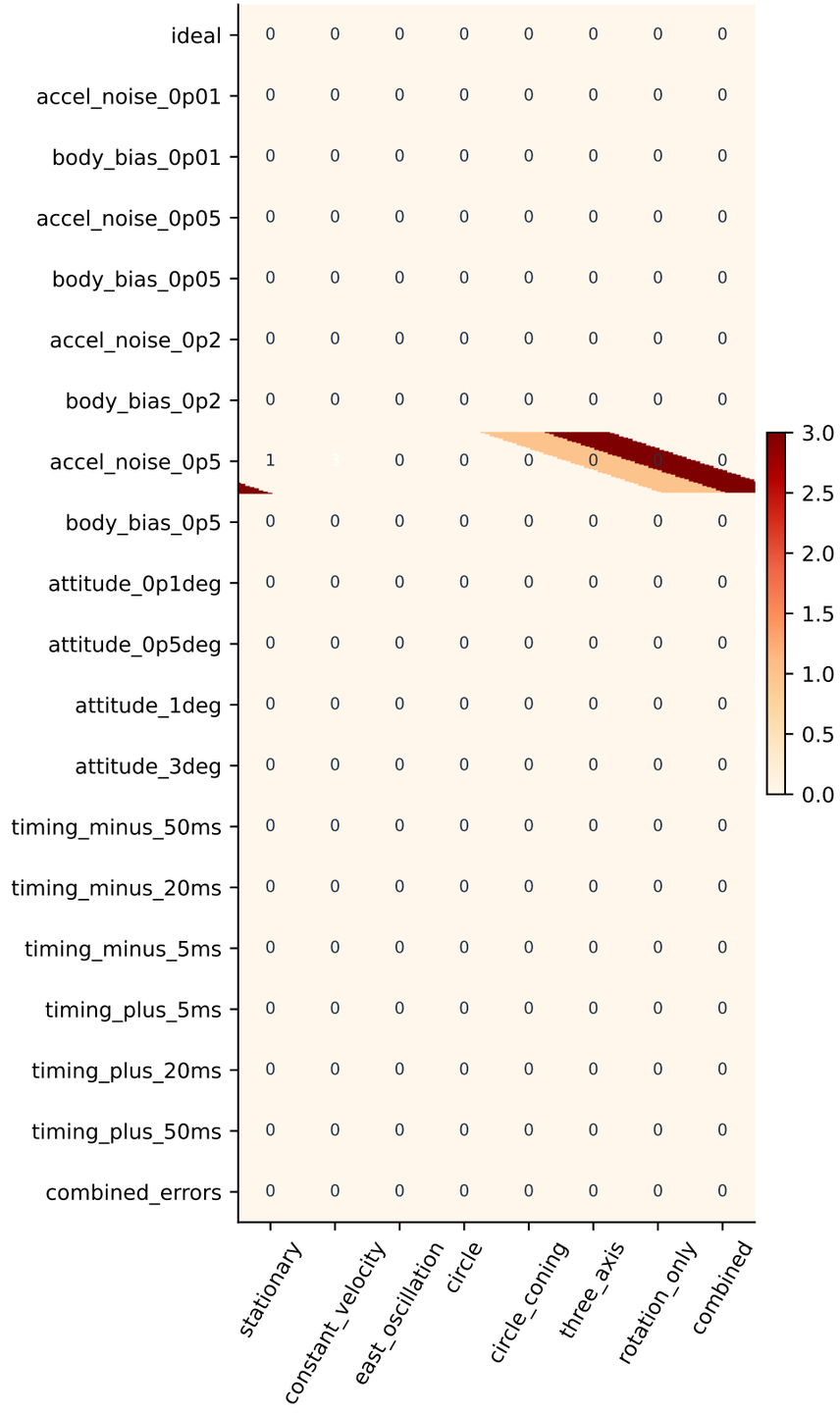
Own IMU integration: controlled-error sensitivity
Inconclusive windows remain separate from wrong decisions; engineering levels, one seed

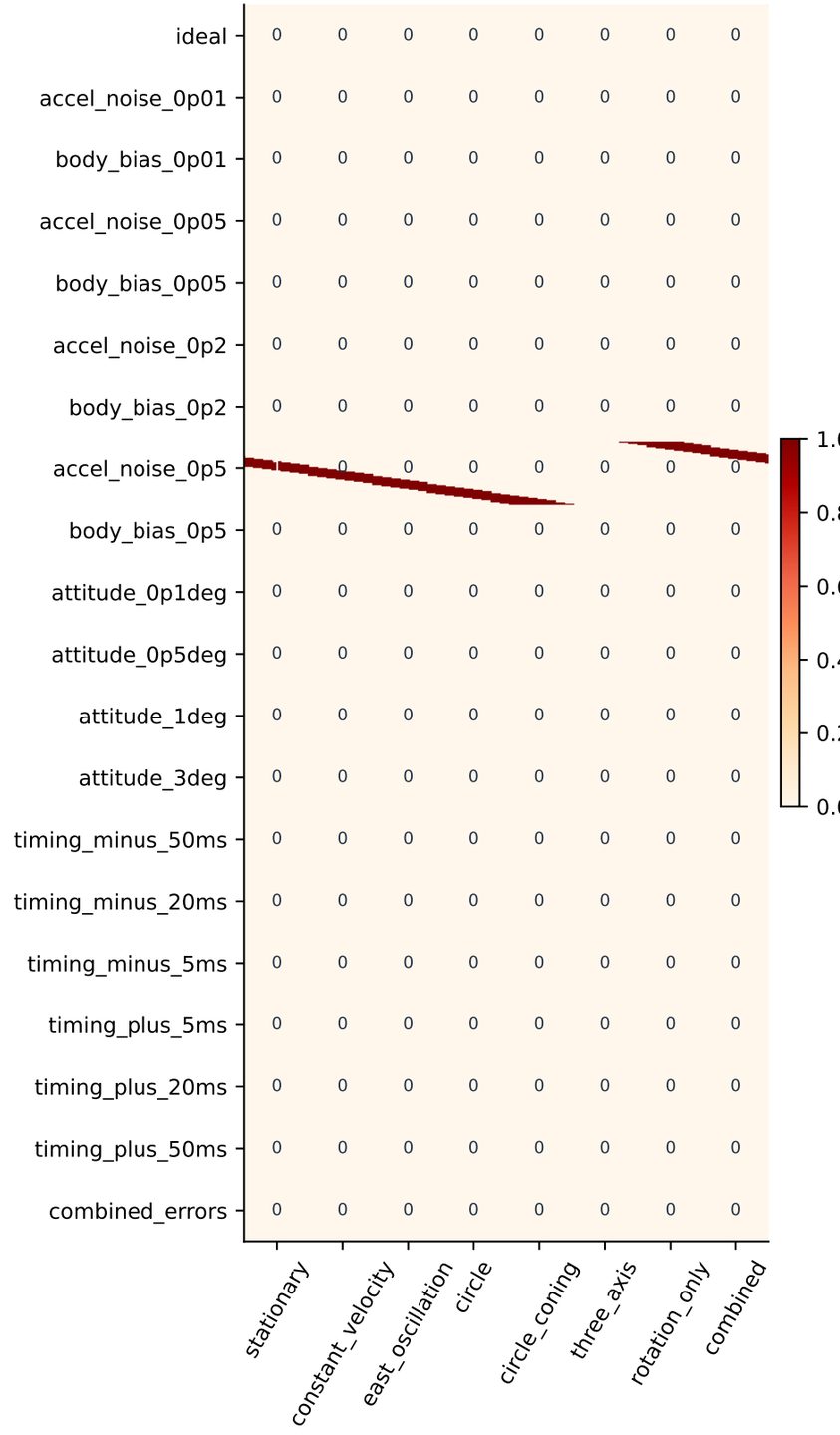
Conclusive coverage (%)

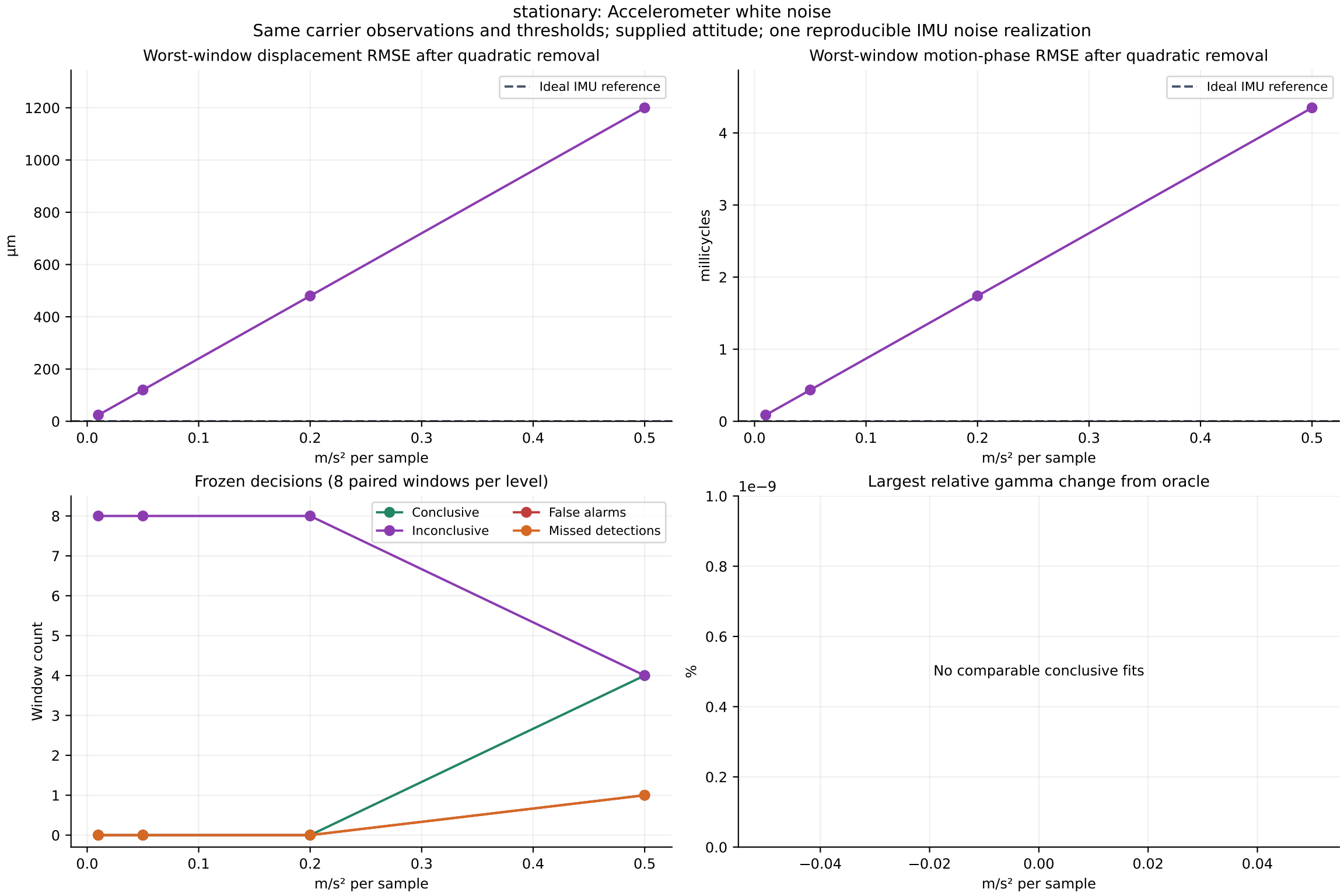


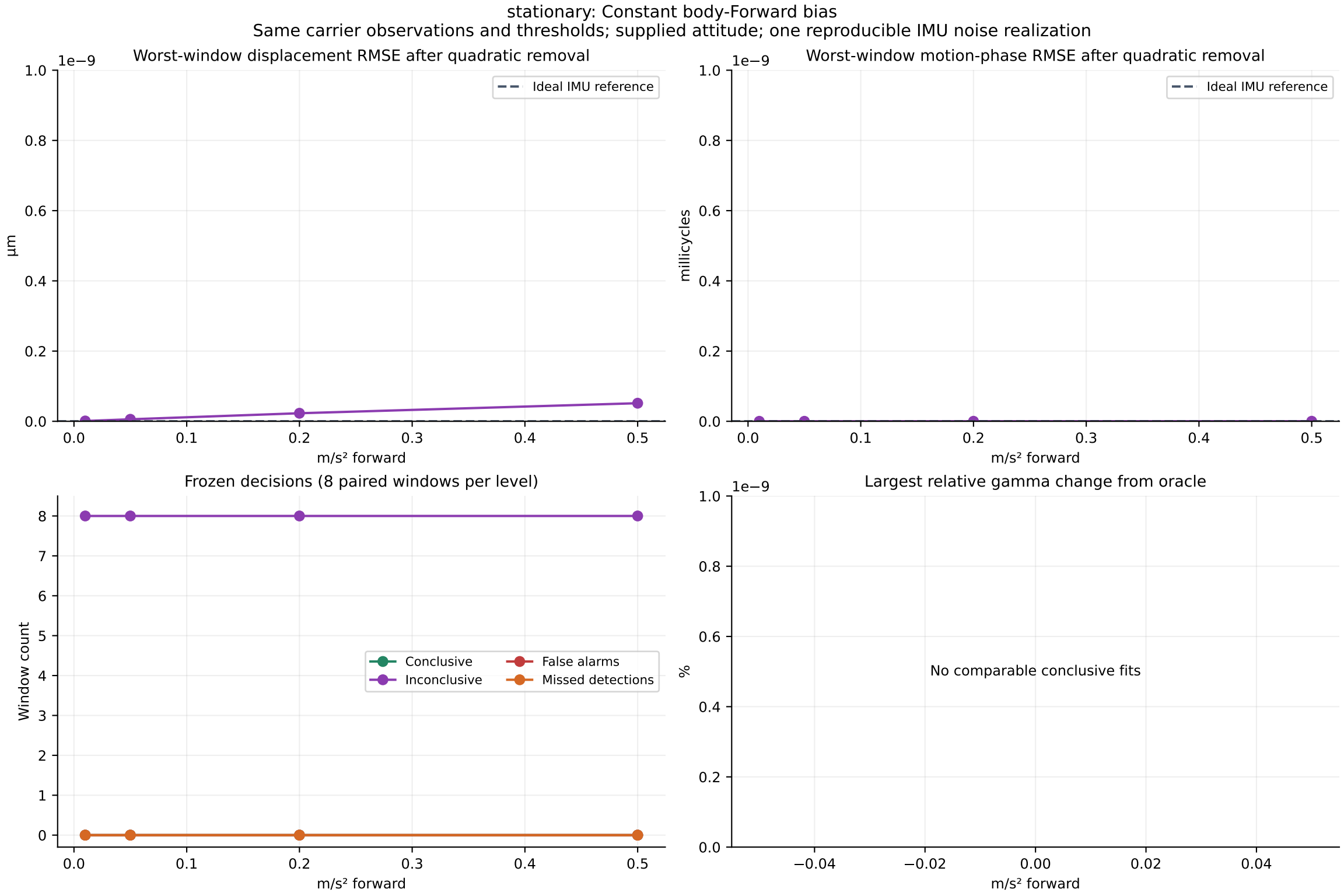
False alarms (window count)

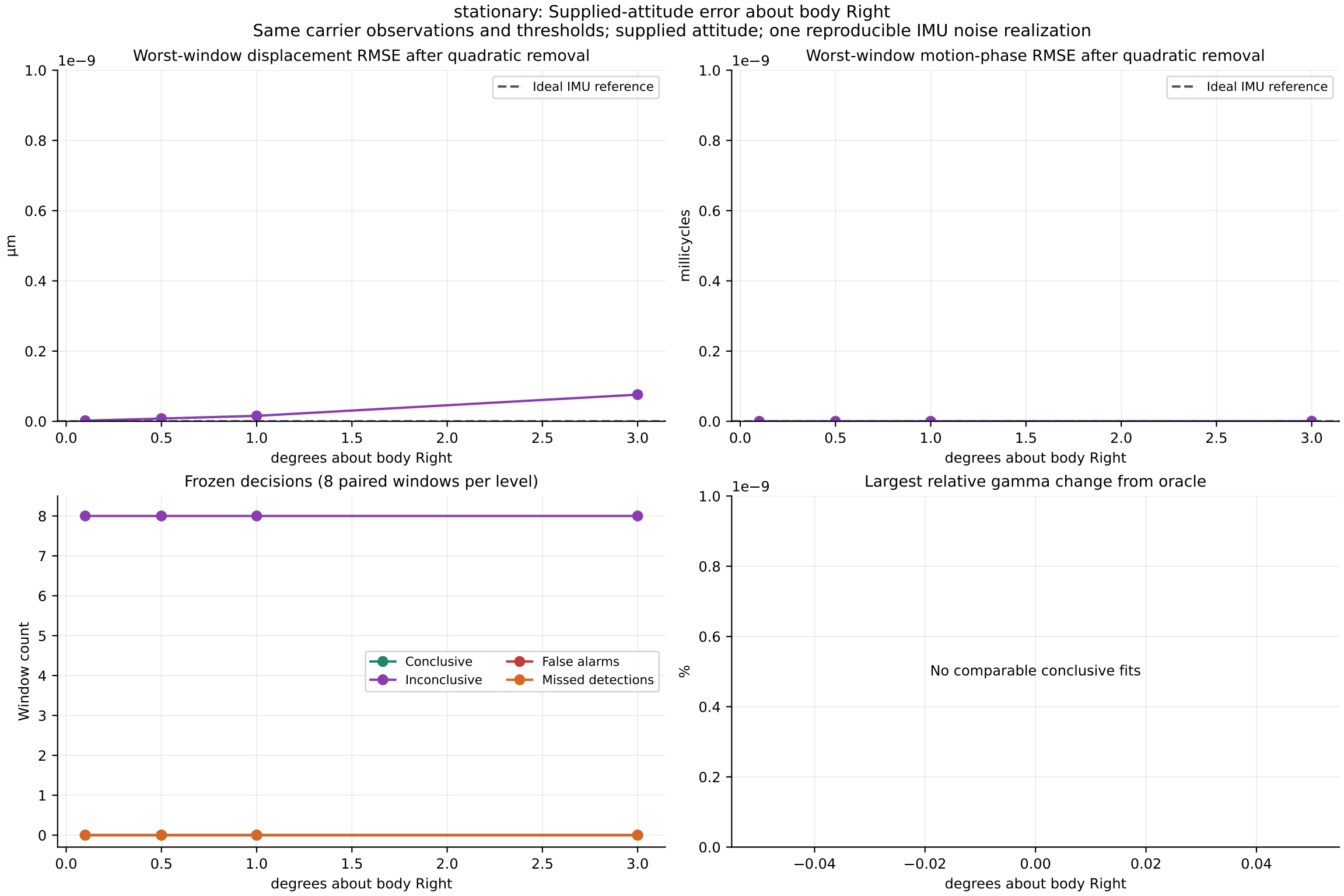


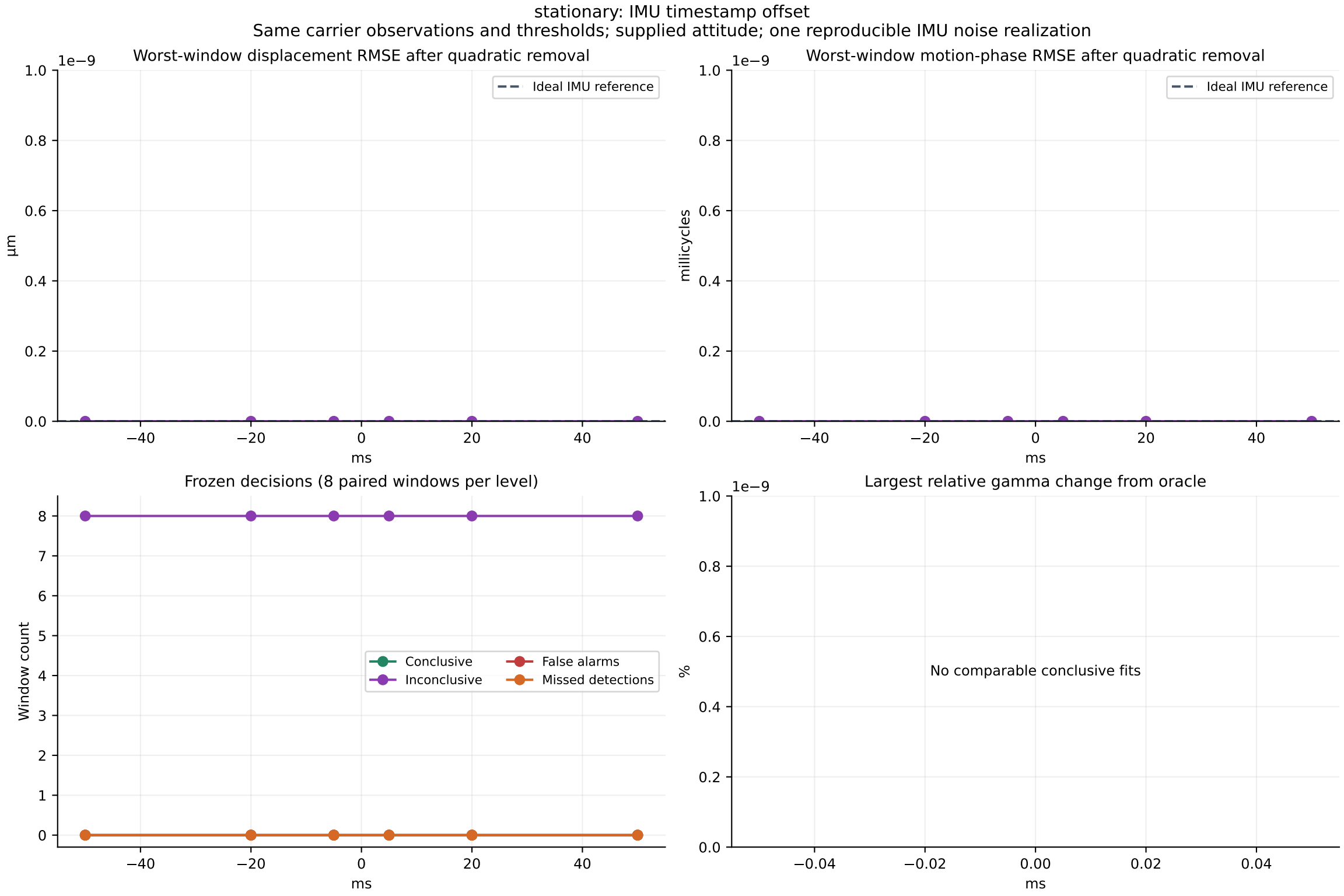
Missed detections (window count)

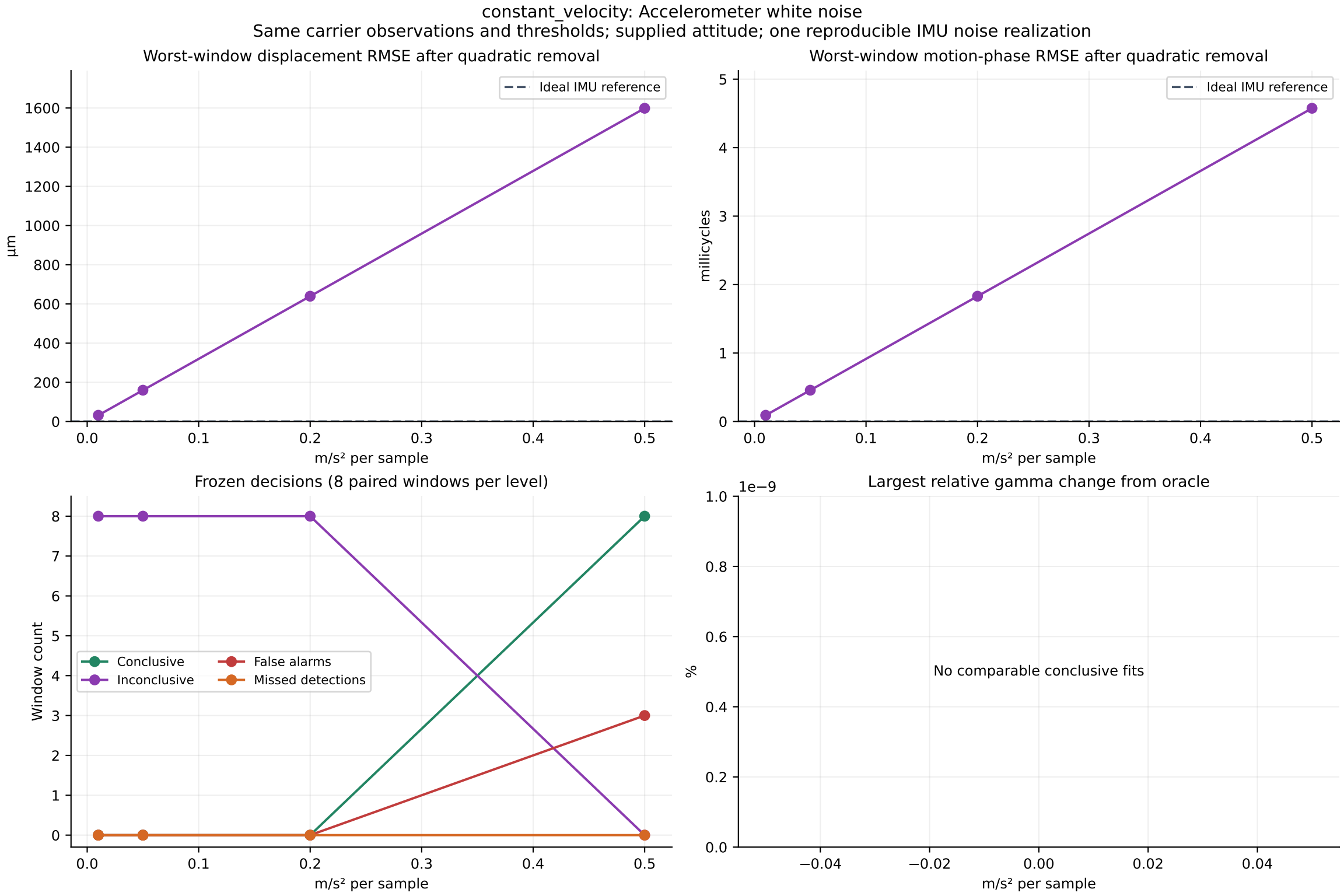


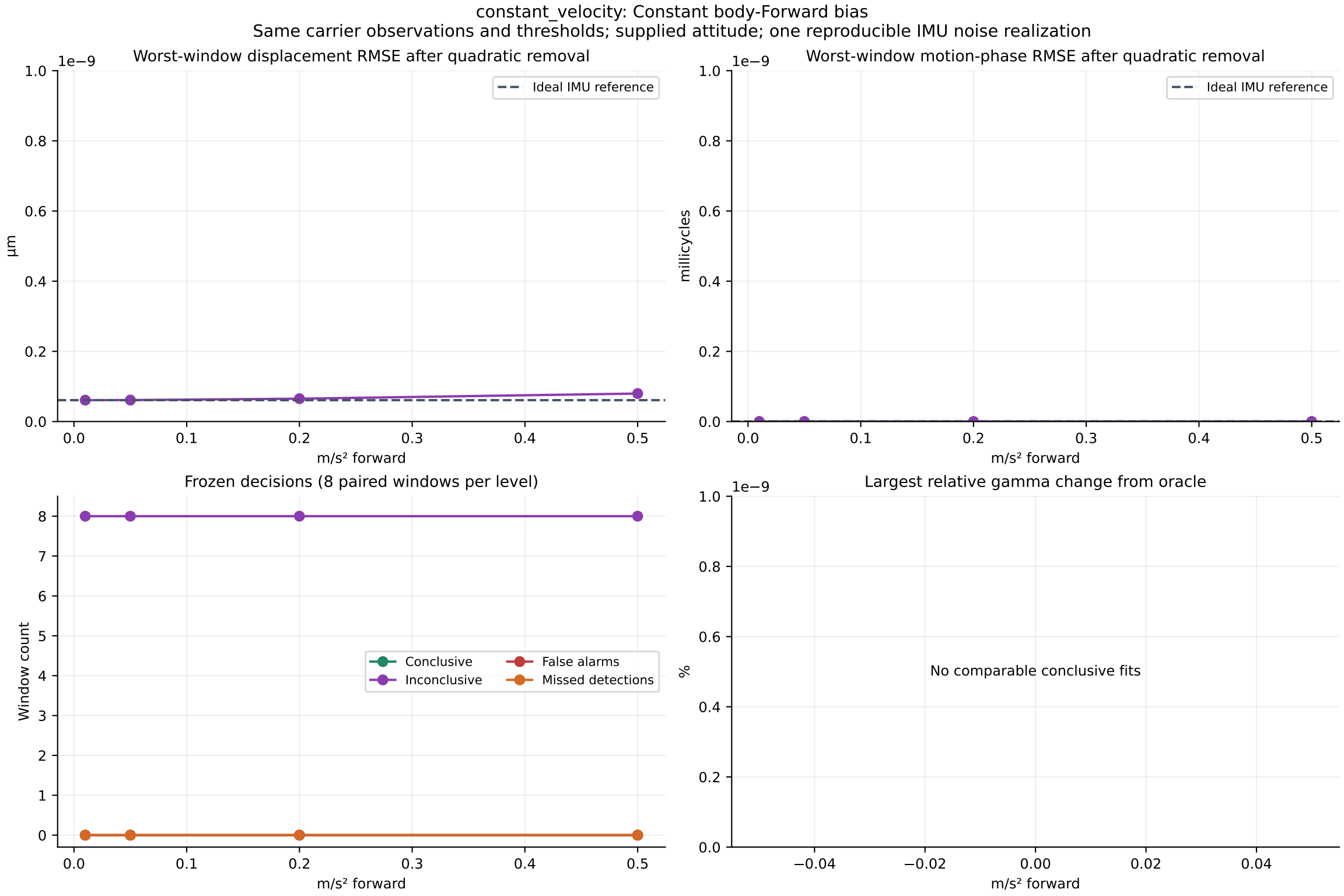

stationary: Accelerometer white noise
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² per sample
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
m/s² per sample
Frozen decisions (8 paired windows per level)
Conclusive
Inconclusive
False alarms
Missed detections
Window count
m/s² per sample
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
m/s² per sample

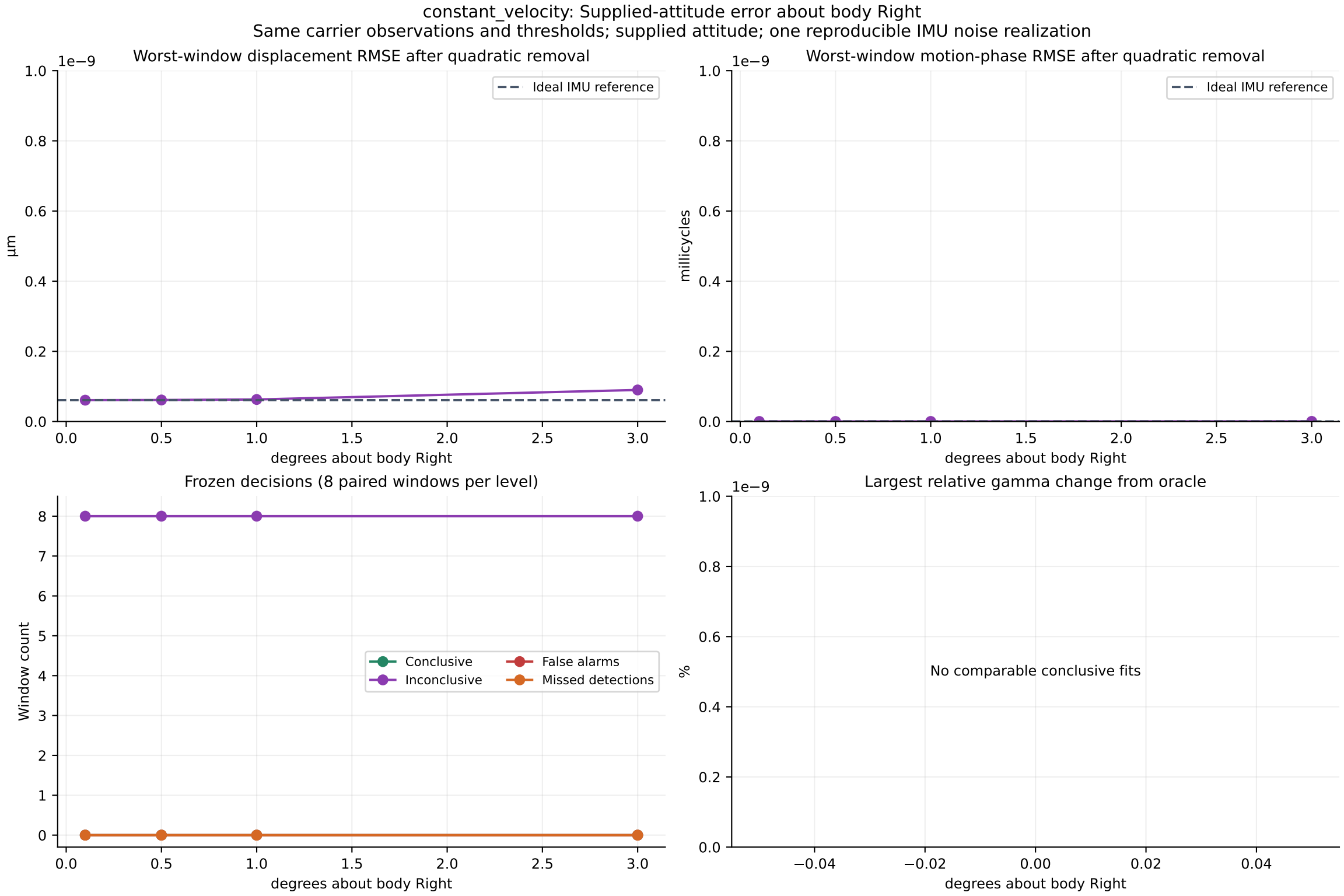

stationary: Constant body-Forward bias
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
1e−9
µm
Ideal IMU reference
m/s² forward
Worst-window motion-phase RMSE after quadratic removal
1e−9
millicycles
Ideal IMU reference
m/s² forward
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
m/s² forward
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
m/s² forward

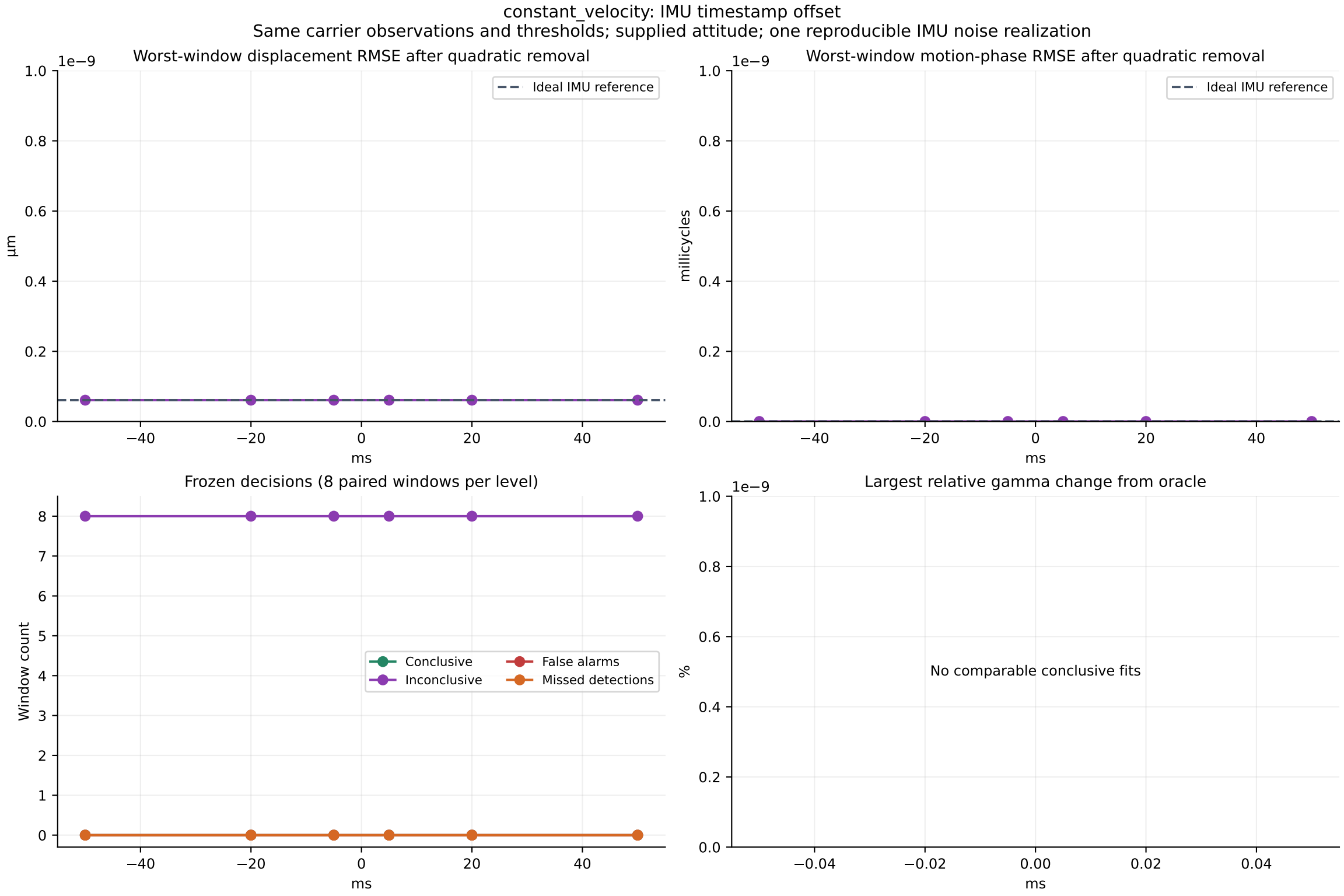

stationary: Supplied-attitude error about body Right
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
1e−9
µm
Ideal IMU reference
degrees about body Right
Worst-window motion-phase RMSE after quadratic removal
1e−9
millicycles
Ideal IMU reference
degrees about body Right
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
degrees about body Right
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
degrees about body Right

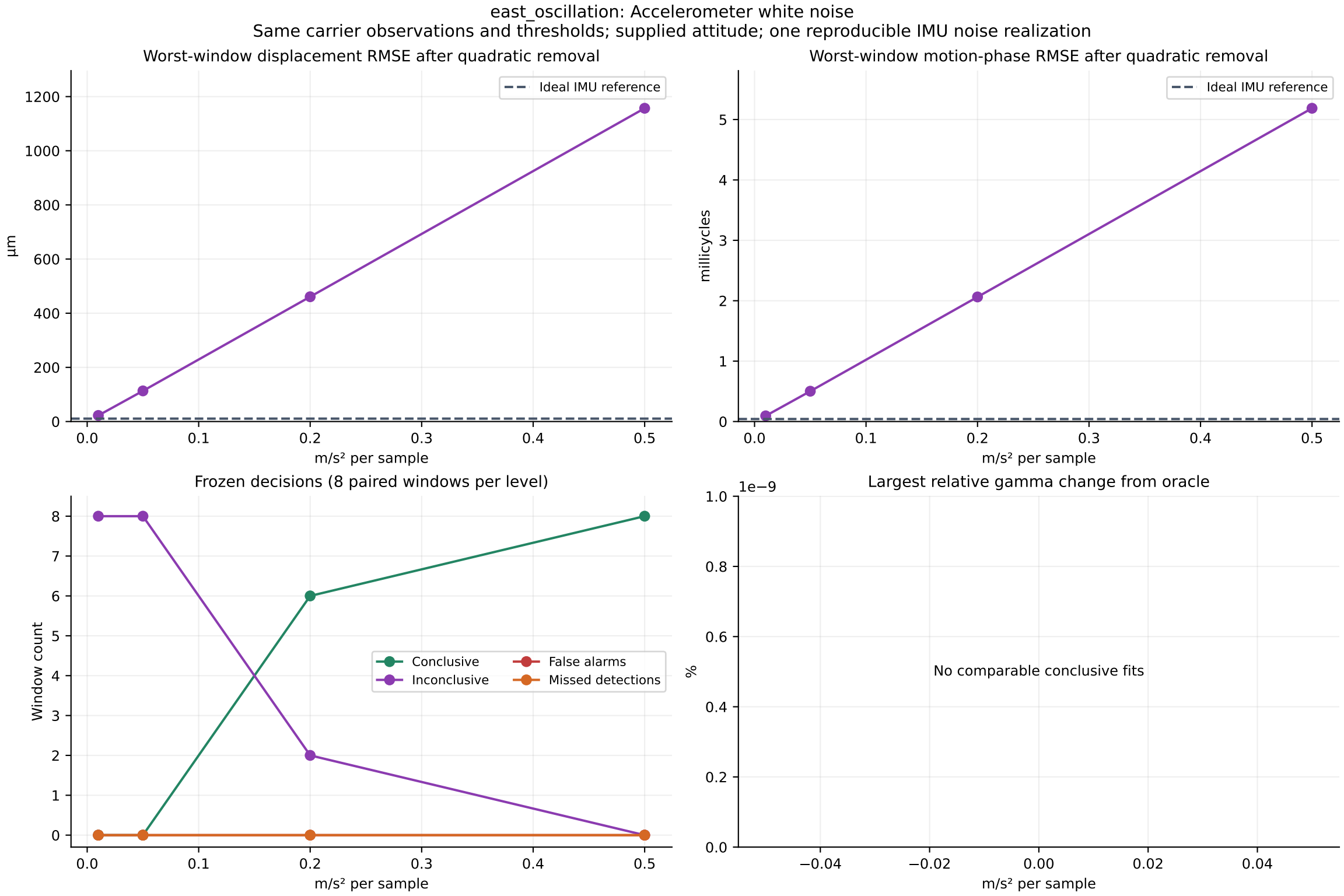

stationary: IMU timestamp offset
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
1e−9
µm
Ideal IMU reference
ms
Worst-window motion-phase RMSE after quadratic removal
millicycles
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%
No comparable conclusive fits

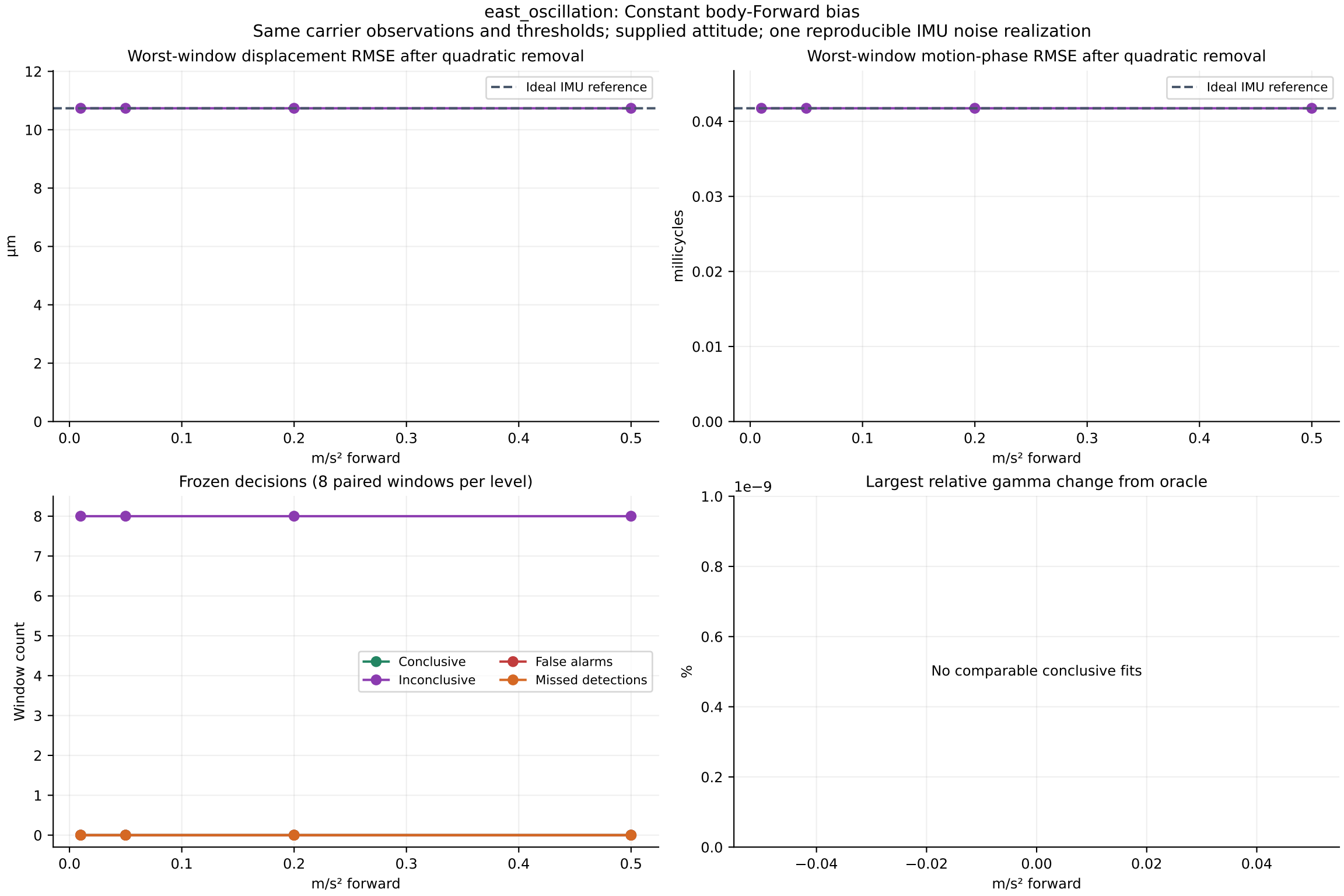

constant_velocity: Accelerometer white noise
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² per sample
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
m/s² per sample
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
m/s² per sample
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
m/s² per sample

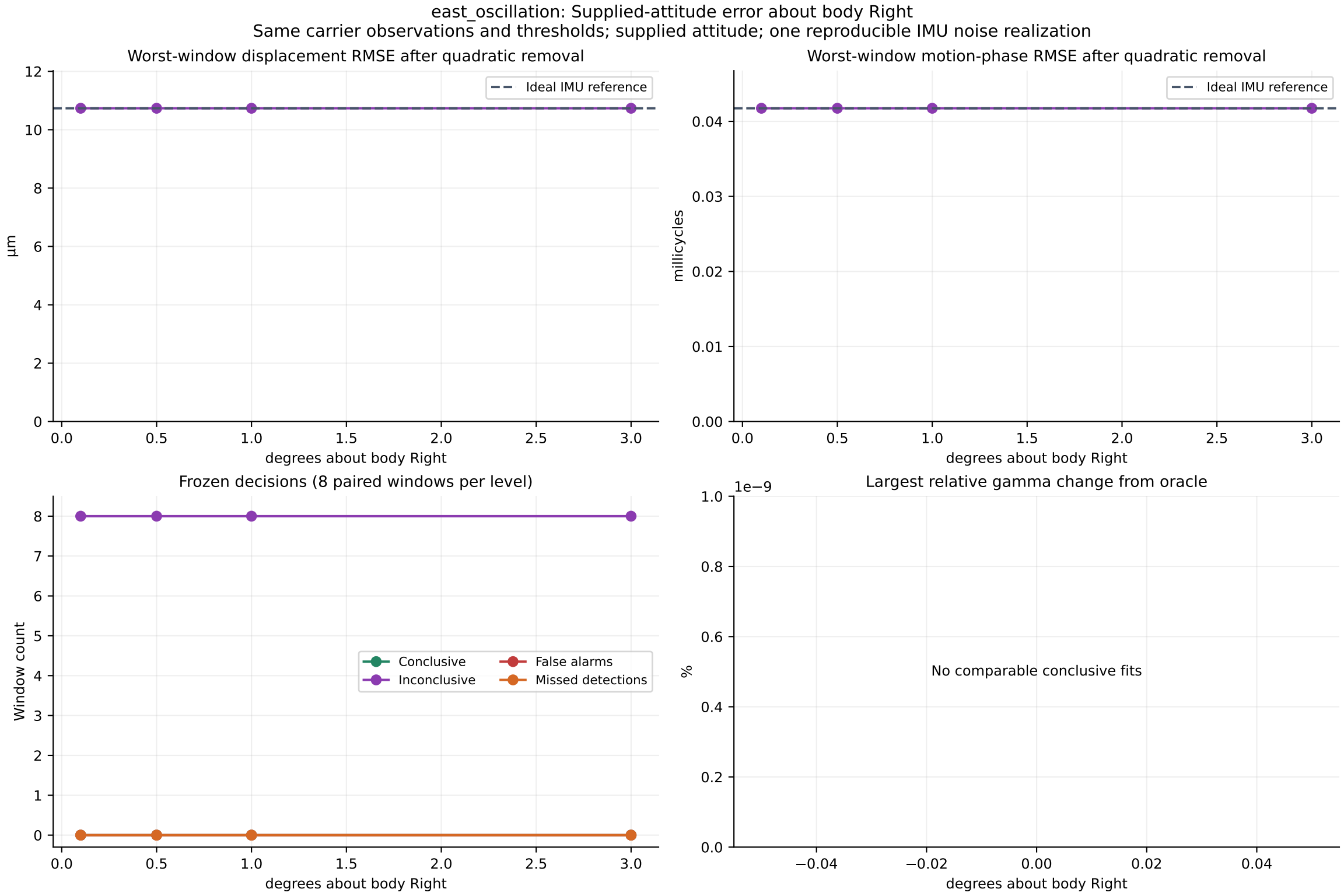

constant_velocity: Constant body-Forward bias
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
1e−9
µm
Ideal IMU reference
m/s² forward
Worst-window motion-phase RMSE after quadratic removal
1e−9
millicycles
Ideal IMU reference
m/s² forward
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
m/s² forward
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
m/s² forward

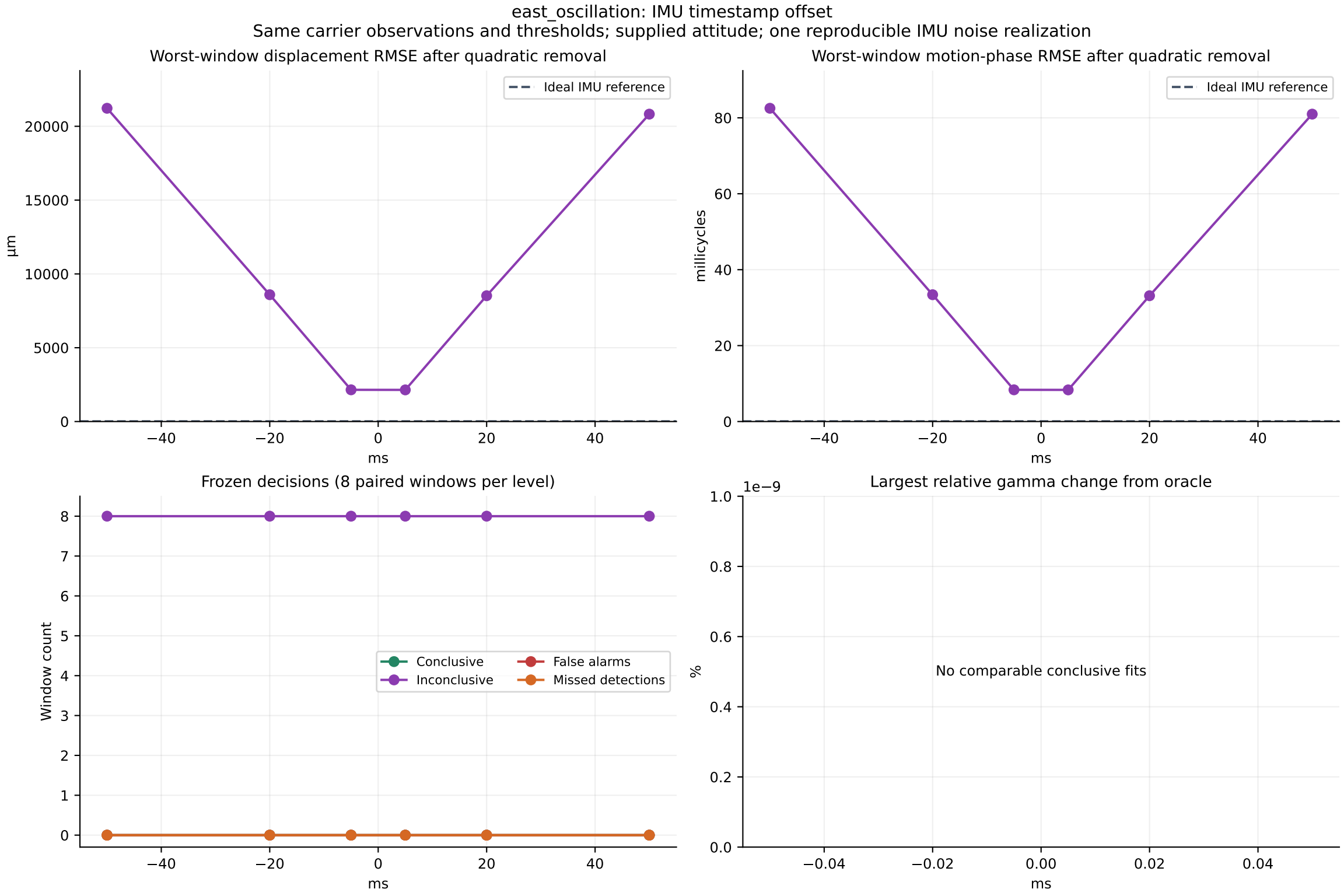

constant_velocity: Supplied-attitude error about body Right
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
1e−9
µm
Ideal IMU reference
degrees about body Right
Worst-window motion-phase RMSE after quadratic removal
1e−9
millicycles
Ideal IMU reference
degrees about body Right
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
degrees about body Right
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
degrees about body Right

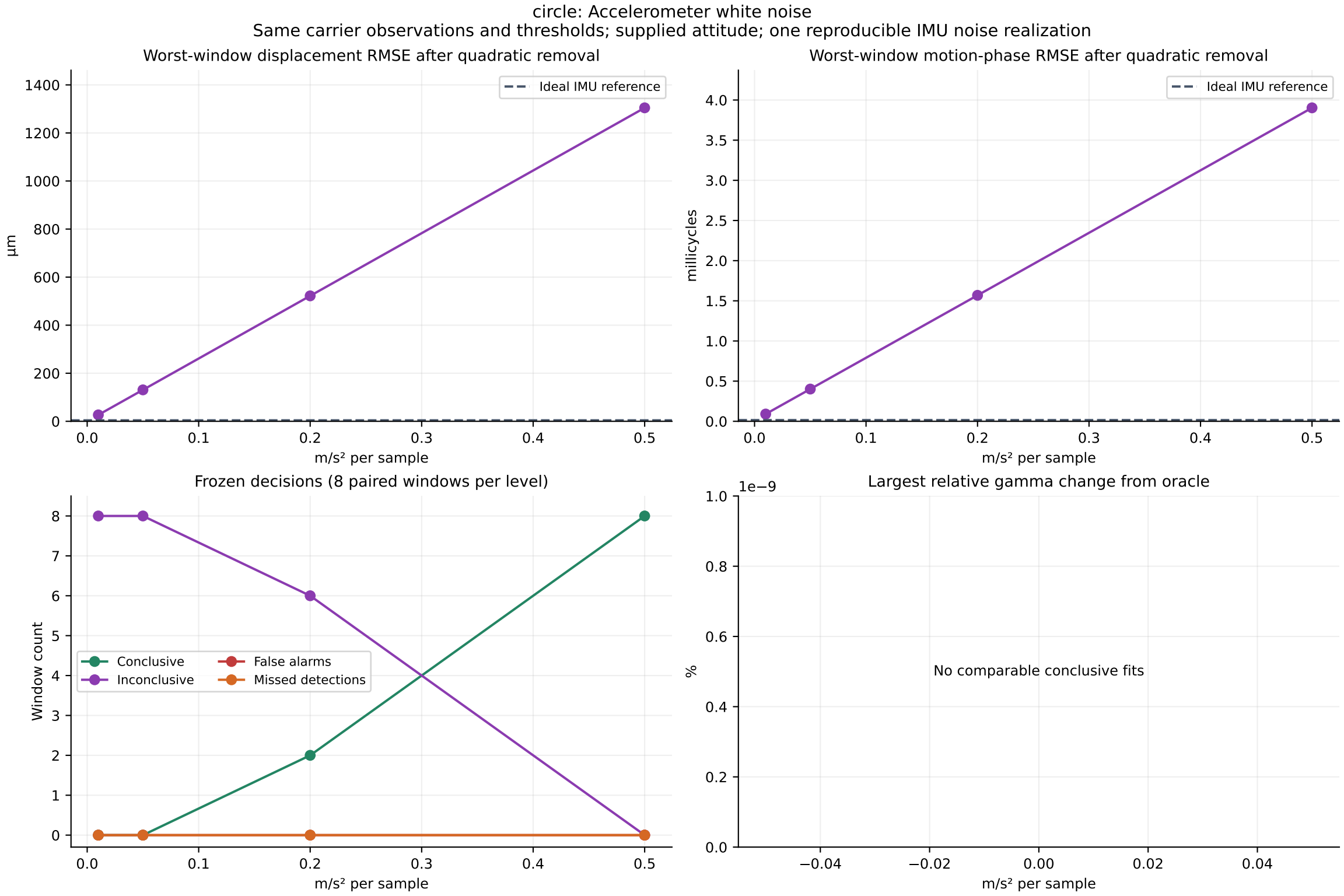

constant_velocity: IMU timestamp offset
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
1e−9
µm
Ideal IMU reference
ms
Worst-window motion-phase RMSE after quadratic removal
millicycles
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%
No comparable conclusive fits

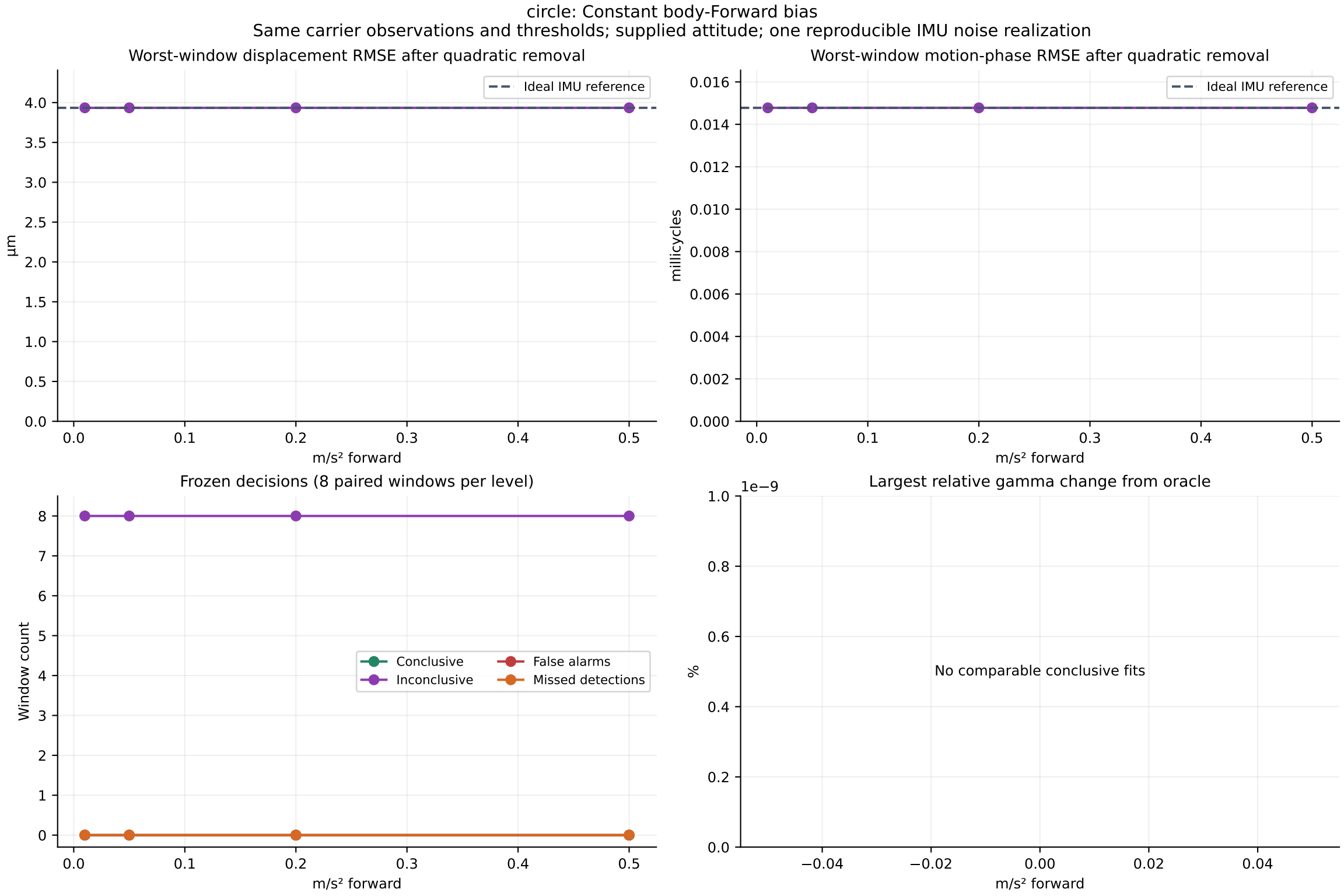

east_oscillation: Accelerometer white noise
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² per sample
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
m/s² per sample
Frozen decisions (8 paired windows per level)
Conclusive
Inconclusive
False alarms
Missed detections
Window count
m/s² per sample
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
m/s² per sample

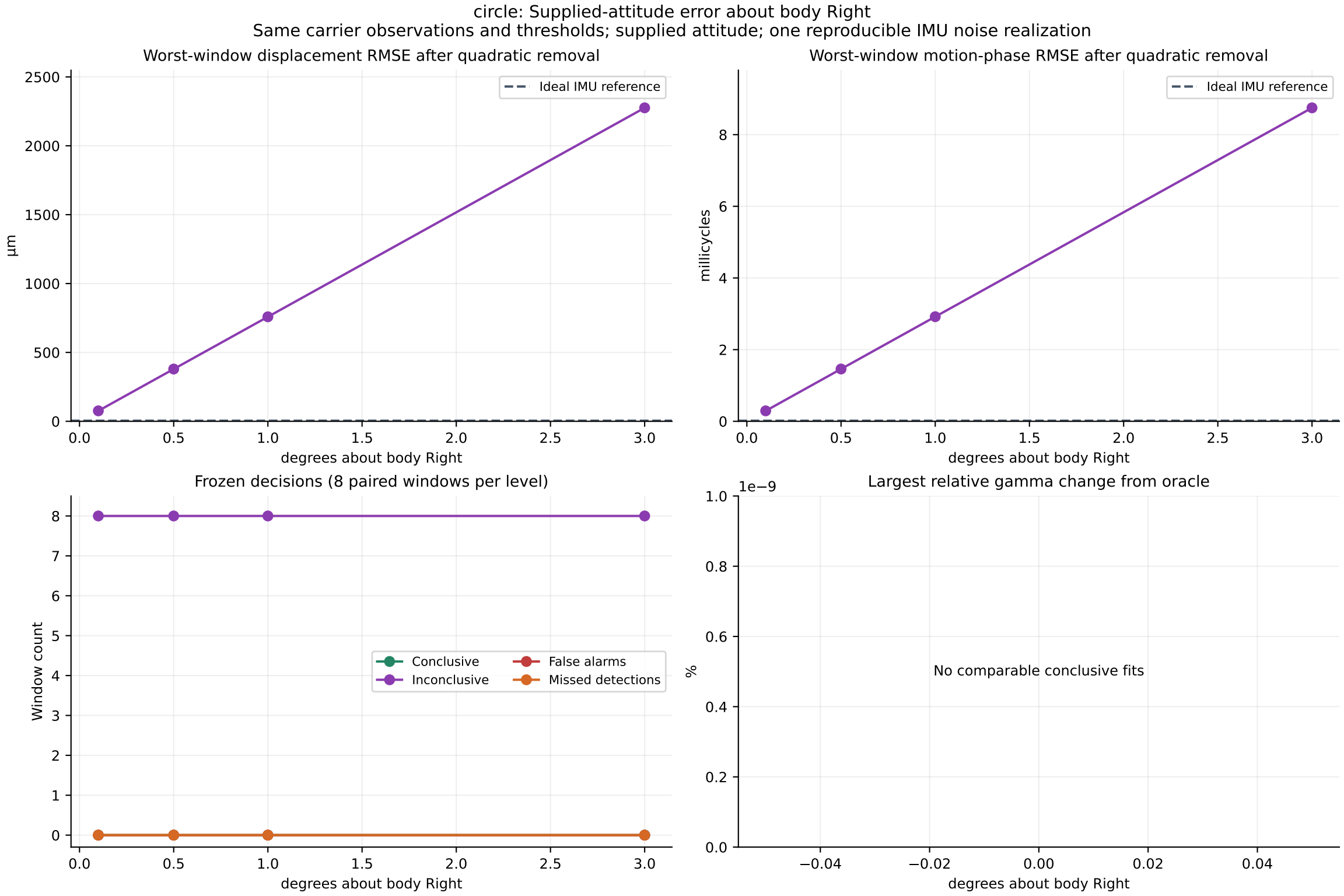

east_oscillation: Constant body-Forward bias
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² forward
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
m/s² forward
Frozen decisions (8 paired windows per level)
Conclusive
Inconclusive
False alarms
Missed detections
Window count
m/s² forward
Largest relative gamma change from oracle
1e−9
No comparable conclusive fits
%
m/s² forward

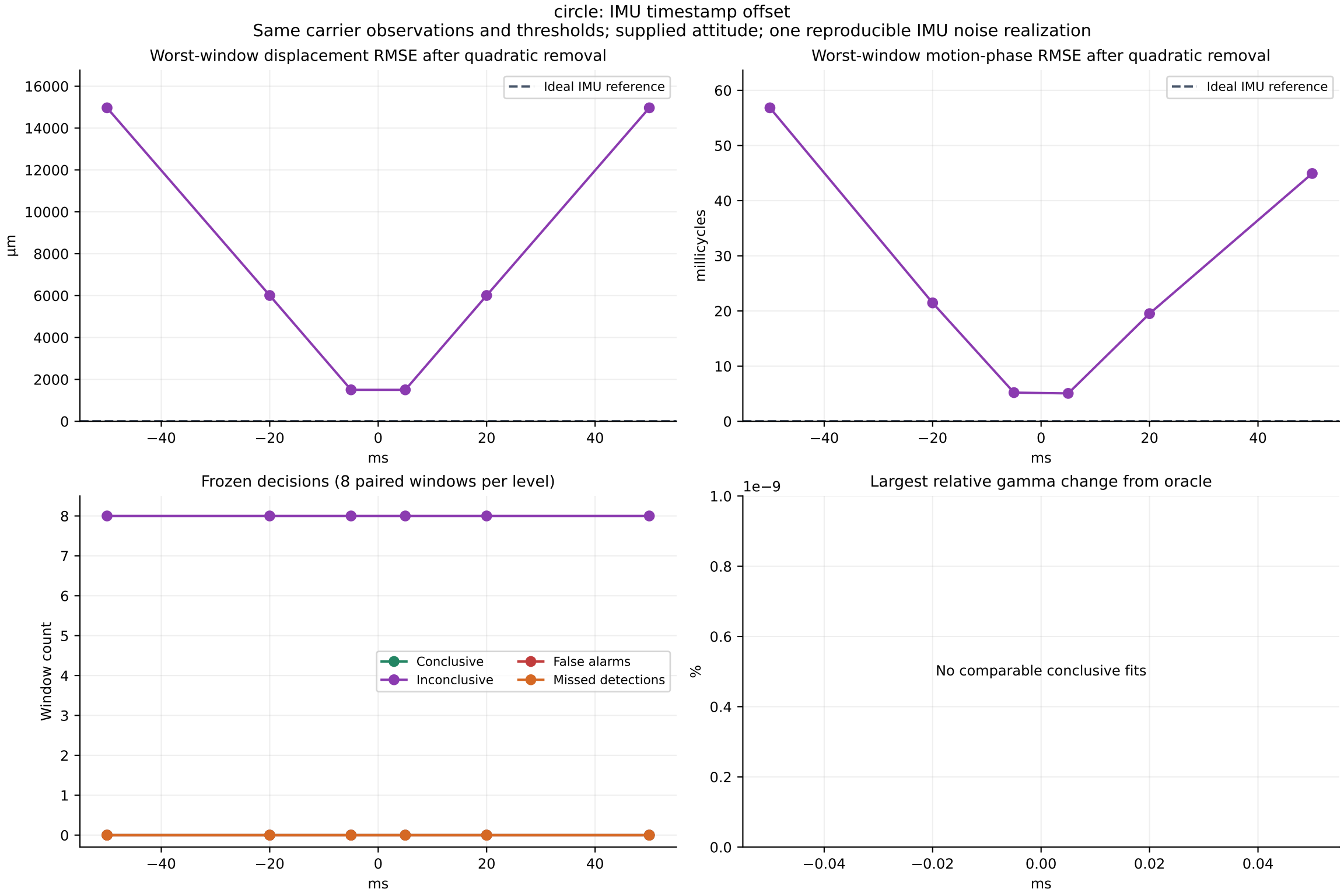

east_oscillation: Supplied-attitude error about body Right
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
degrees about body Right
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
degrees about body Right
Frozen decisions (8 paired windows per level)
Conclusive
Inconclusive
False alarms
Missed detections
Window count
degrees about body Right
Largest relative gamma change from oracle
1e−9
No comparable conclusive fits
%
degrees about body Right

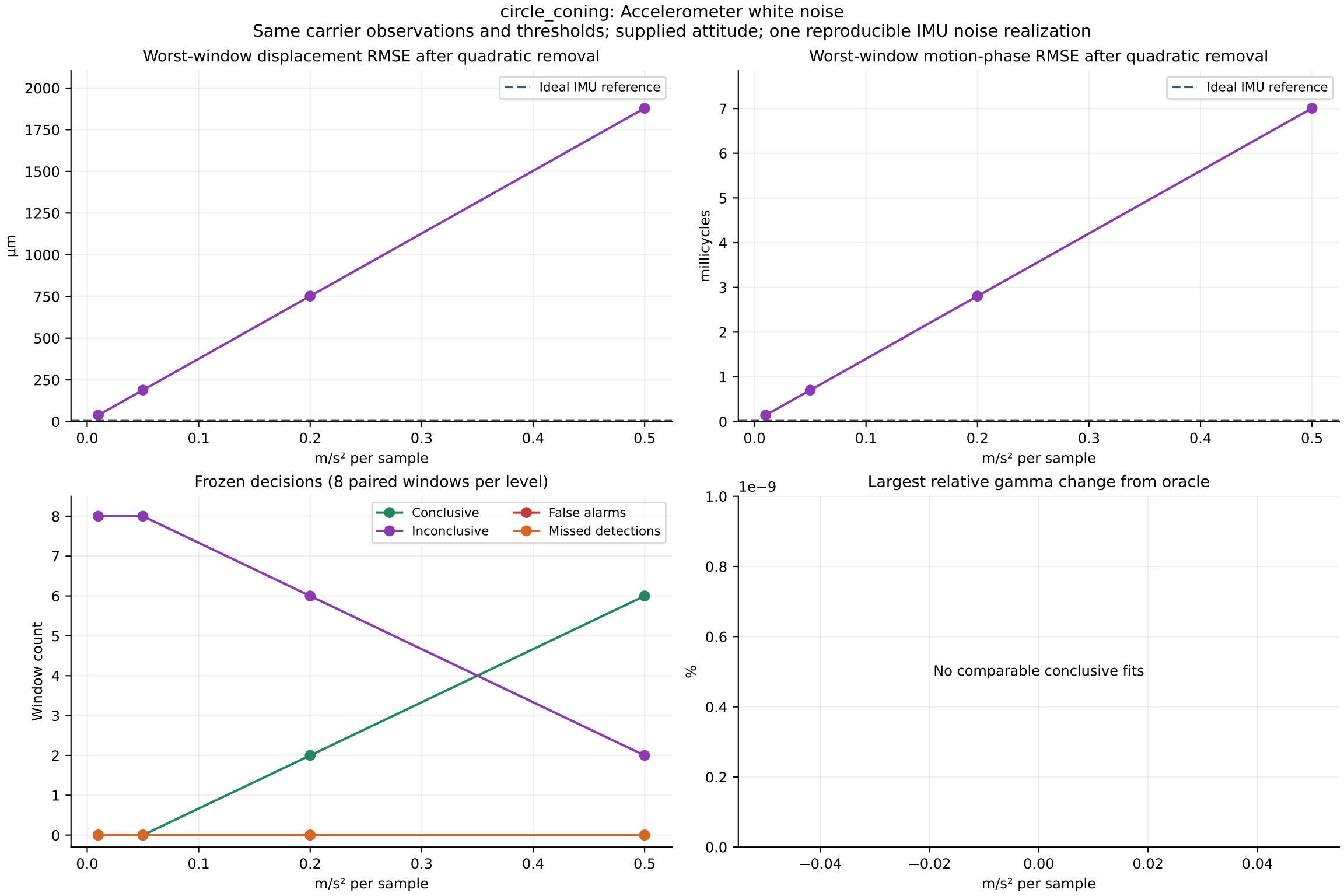

east_oscillation: IMU timestamp offset
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
ms
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
ms
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
ms
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
ms

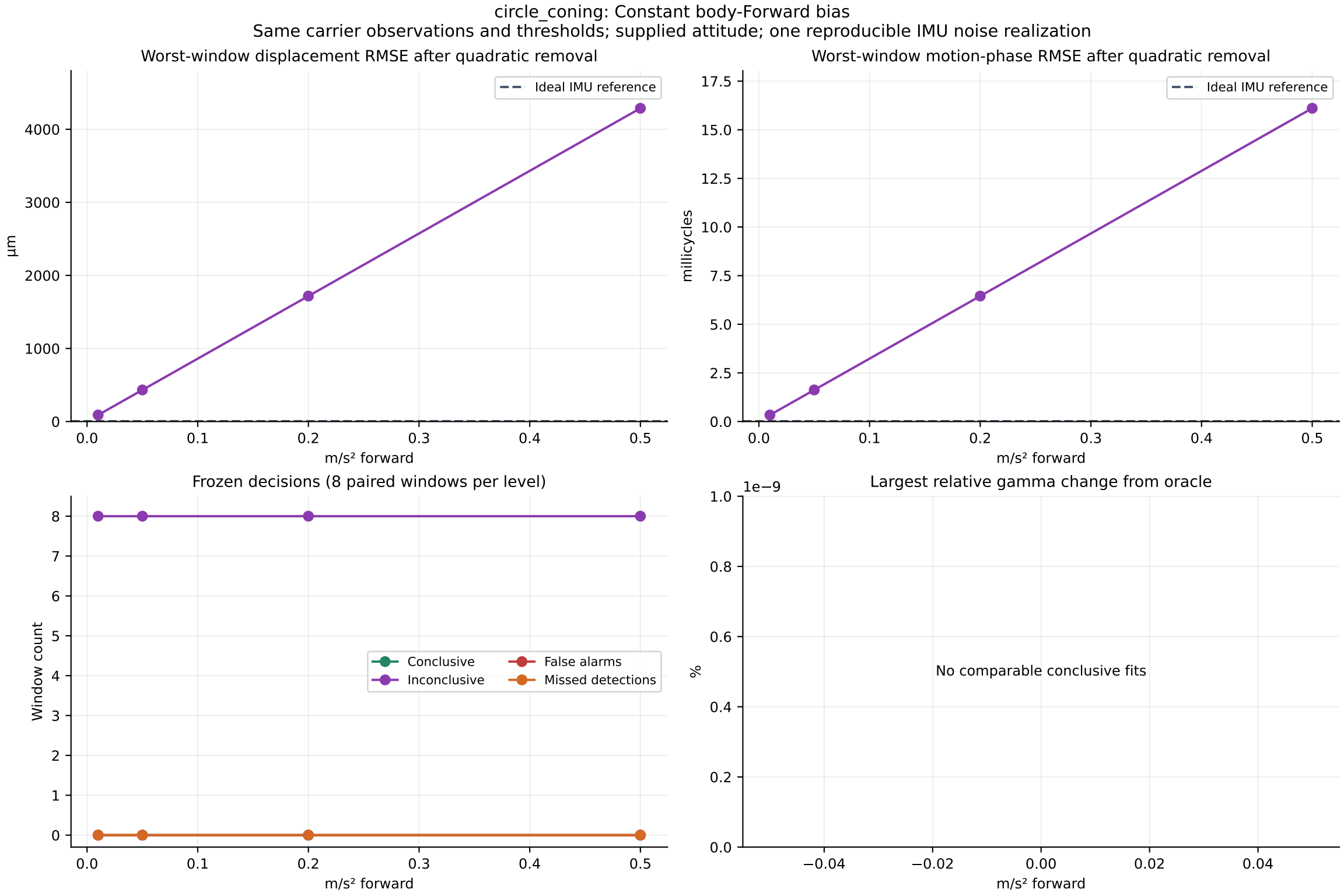

circle: Accelerometer white noise
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² per sample
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
m/s² per sample
Frozen decisions (8 paired windows per level)
Conclusive
Inconclusive
False alarms
Missed detections
Window count
m/s² per sample
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
m/s² per sample

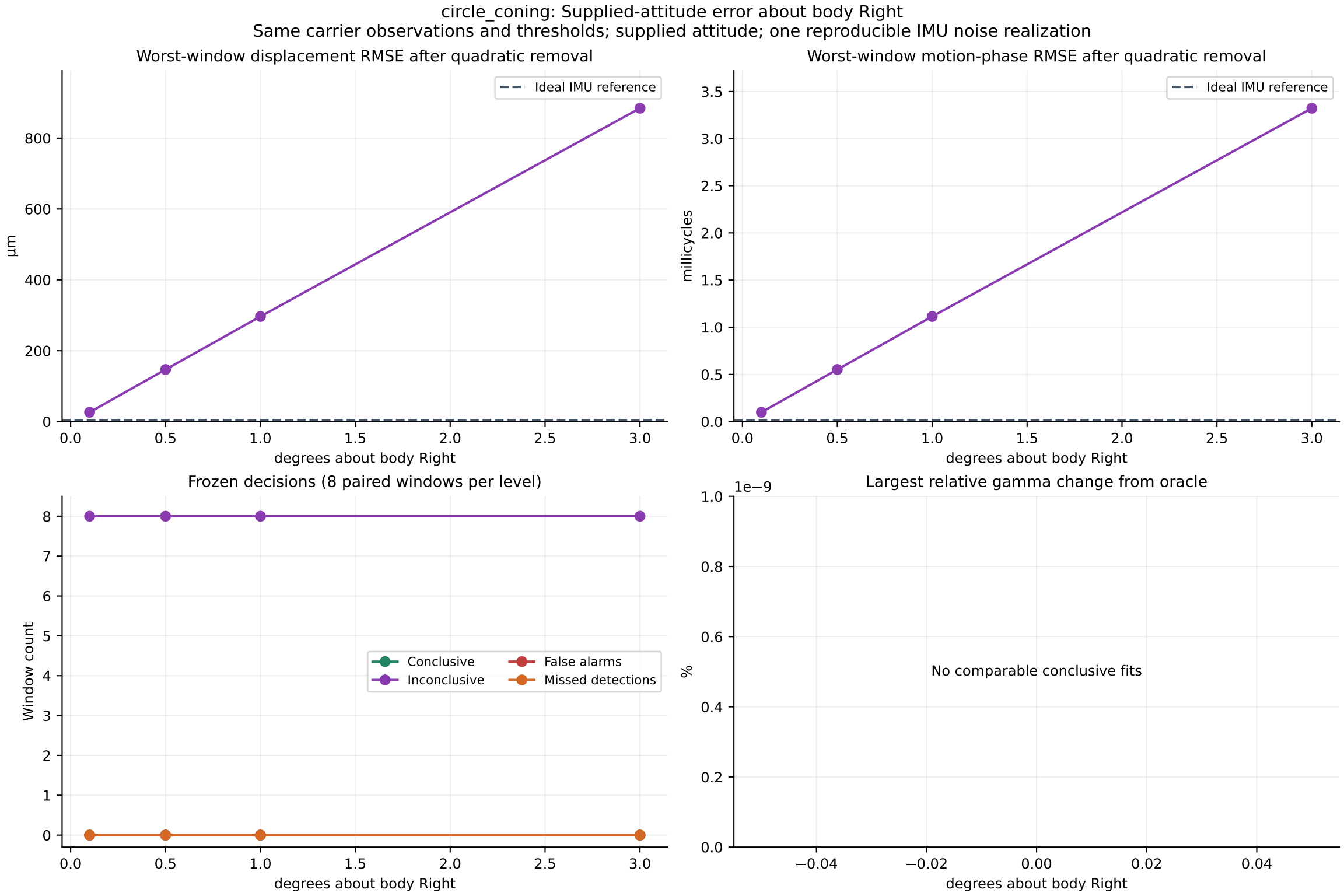

circle: Constant body-Forward bias
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² forward
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
m/s² forward
Frozen decisions (8 paired windows per level)
Conclusive
Inconclusive
False alarms
Missed detections
Window count
m/s² forward
Largest relative gamma change from oracle
1e−9
No comparable conclusive fits
%
m/s² forward

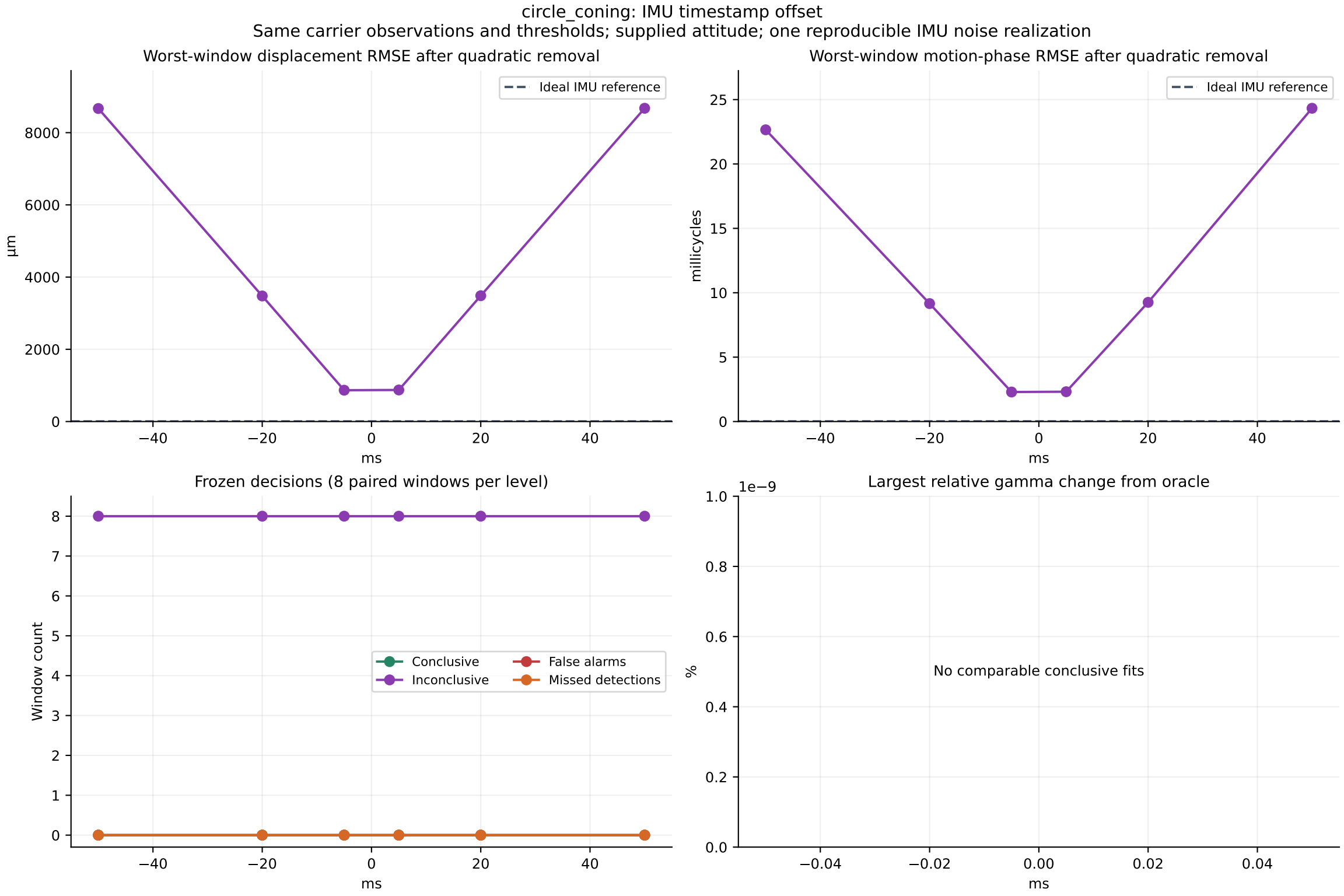

circle: Supplied-attitude error about body Right
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
degrees about body Right
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
degrees about body Right
Frozen decisions (8 paired windows per level)
Conclusive
Inconclusive
False alarms
Missed detections
Window count
degrees about body Right
Largest relative gamma change from oracle
1e−9
No comparable conclusive fits
%
degrees about body Right

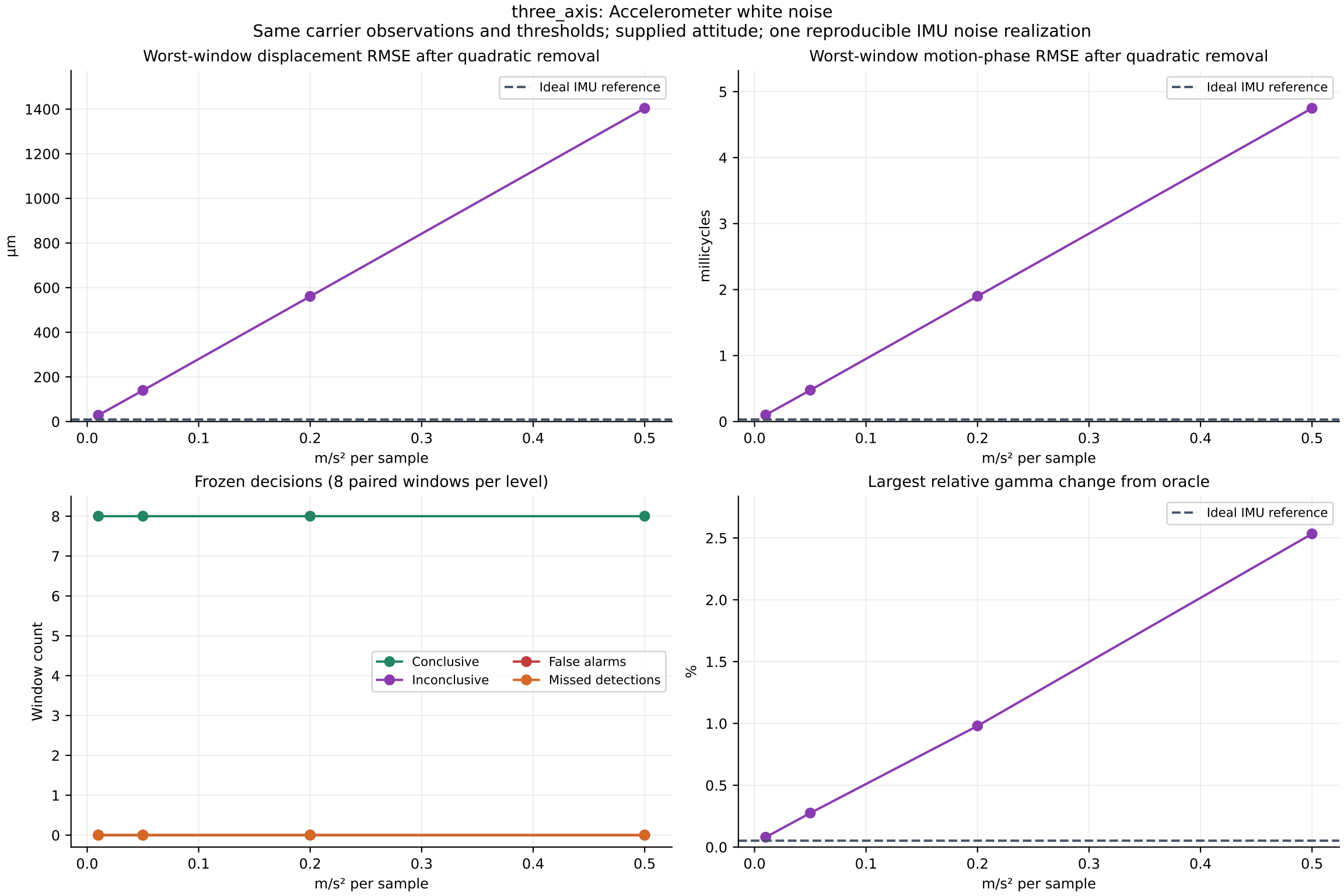

circle: IMU timestamp offset
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
ms
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
ms
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
ms
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
ms

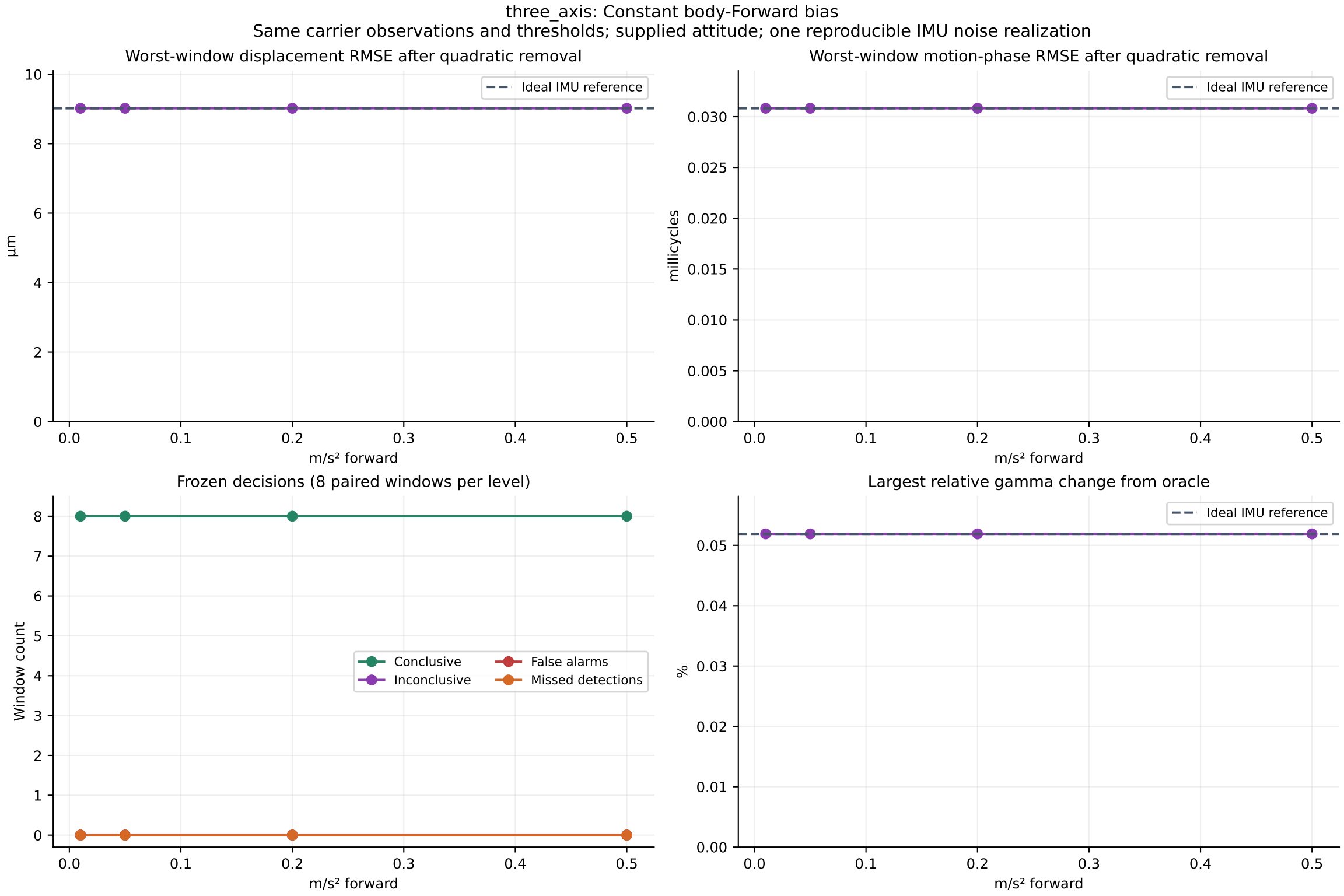

circle_coning: Accelerometer white noise
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² per sample
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
m/s² per sample
Frozen decisions (8 paired windows per level)
Conclusive
Inconclusive
False alarms
Missed detections
Window count
m/s² per sample
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
m/s² per sample

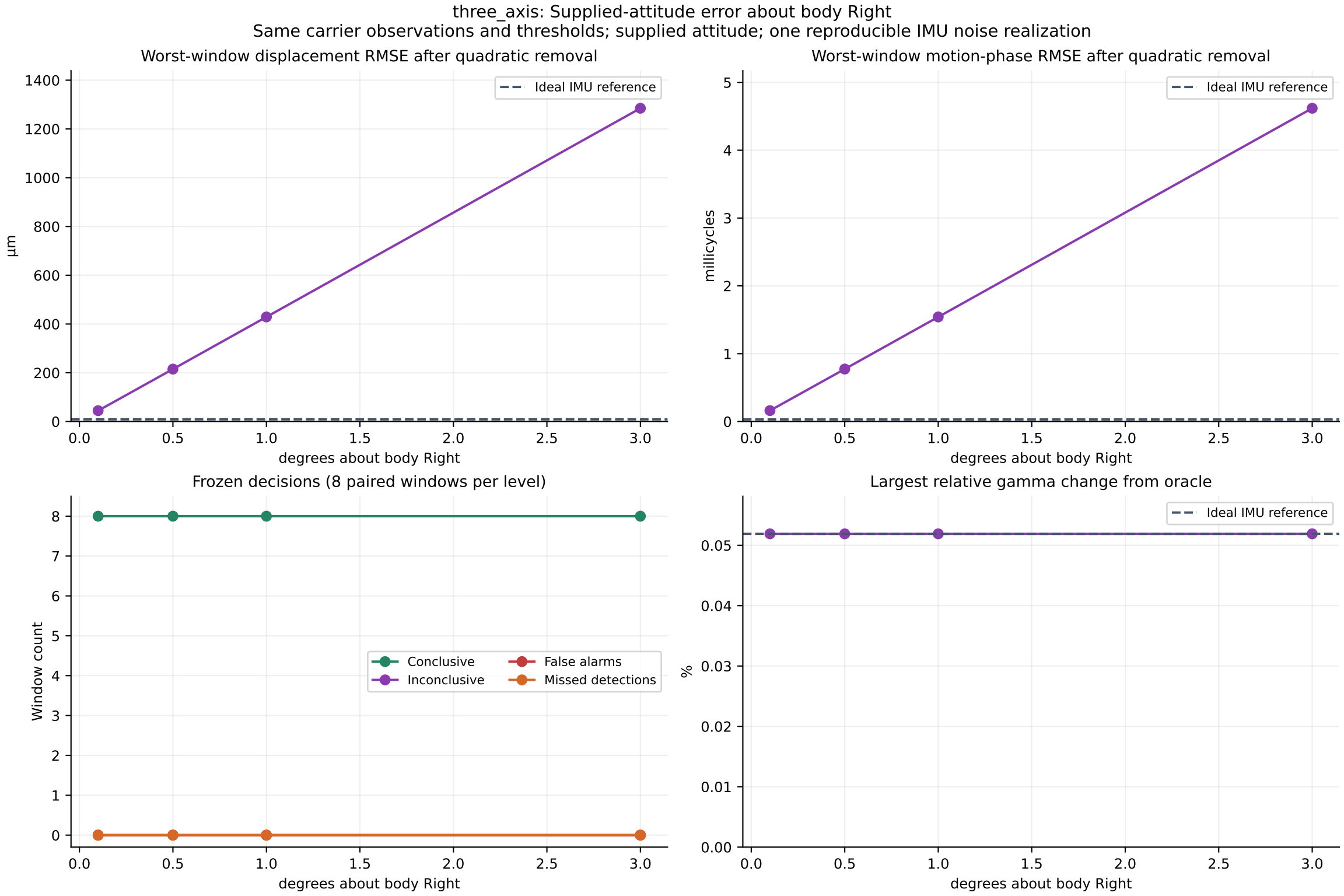

circle_coning: Constant body-Forward bias
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² forward
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
m/s² forward
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
m/s² forward
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
m/s² forward

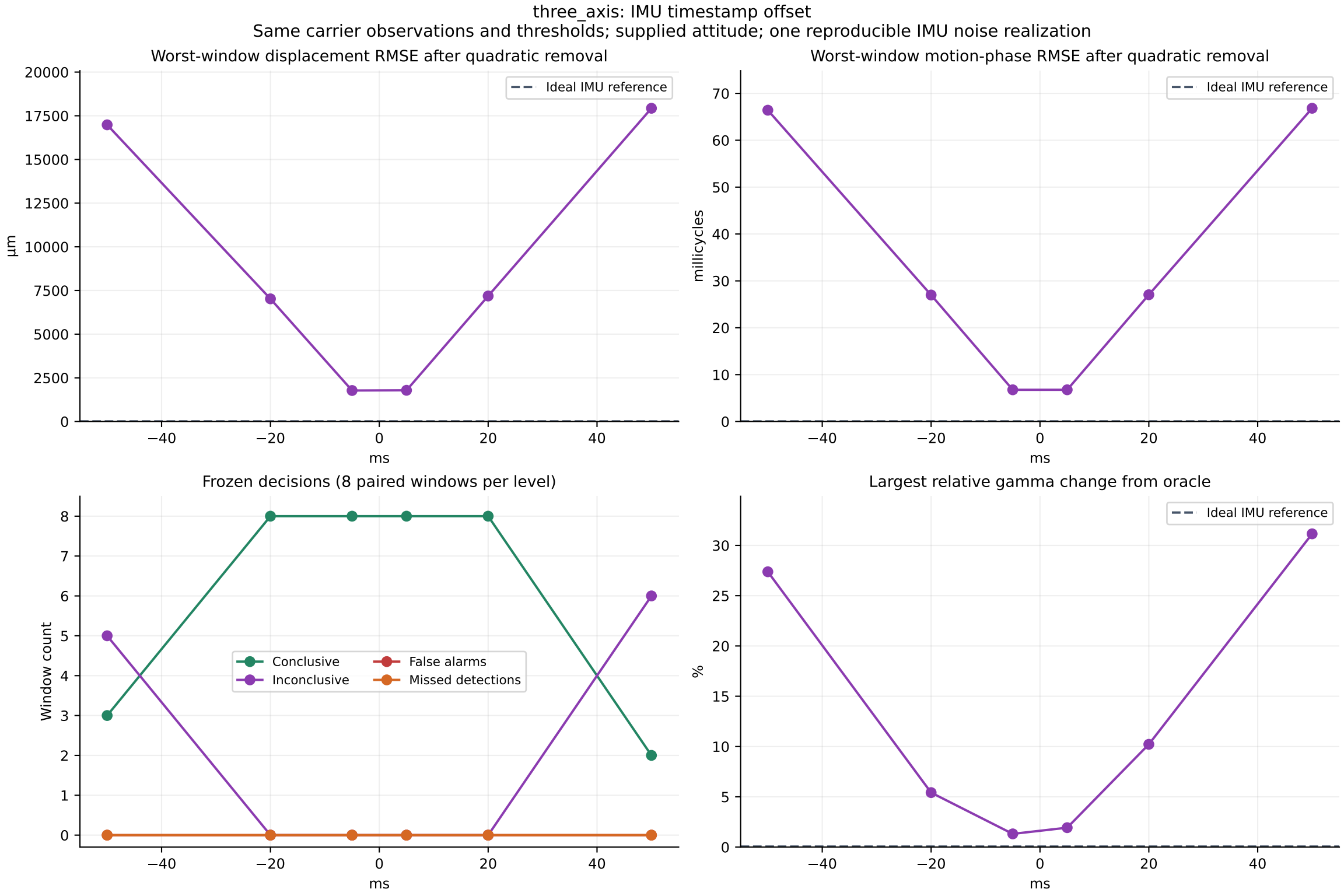

circle_coning: Supplied-attitude error about body Right
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
degrees about body Right
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
degrees about body Right
Frozen decisions (8 paired windows per level)
Conclusive
Inconclusive
False alarms
Missed detections
Window count
degrees about body Right
Largest relative gamma change from oracle
1e−9
No comparable conclusive fits
%
degrees about body Right

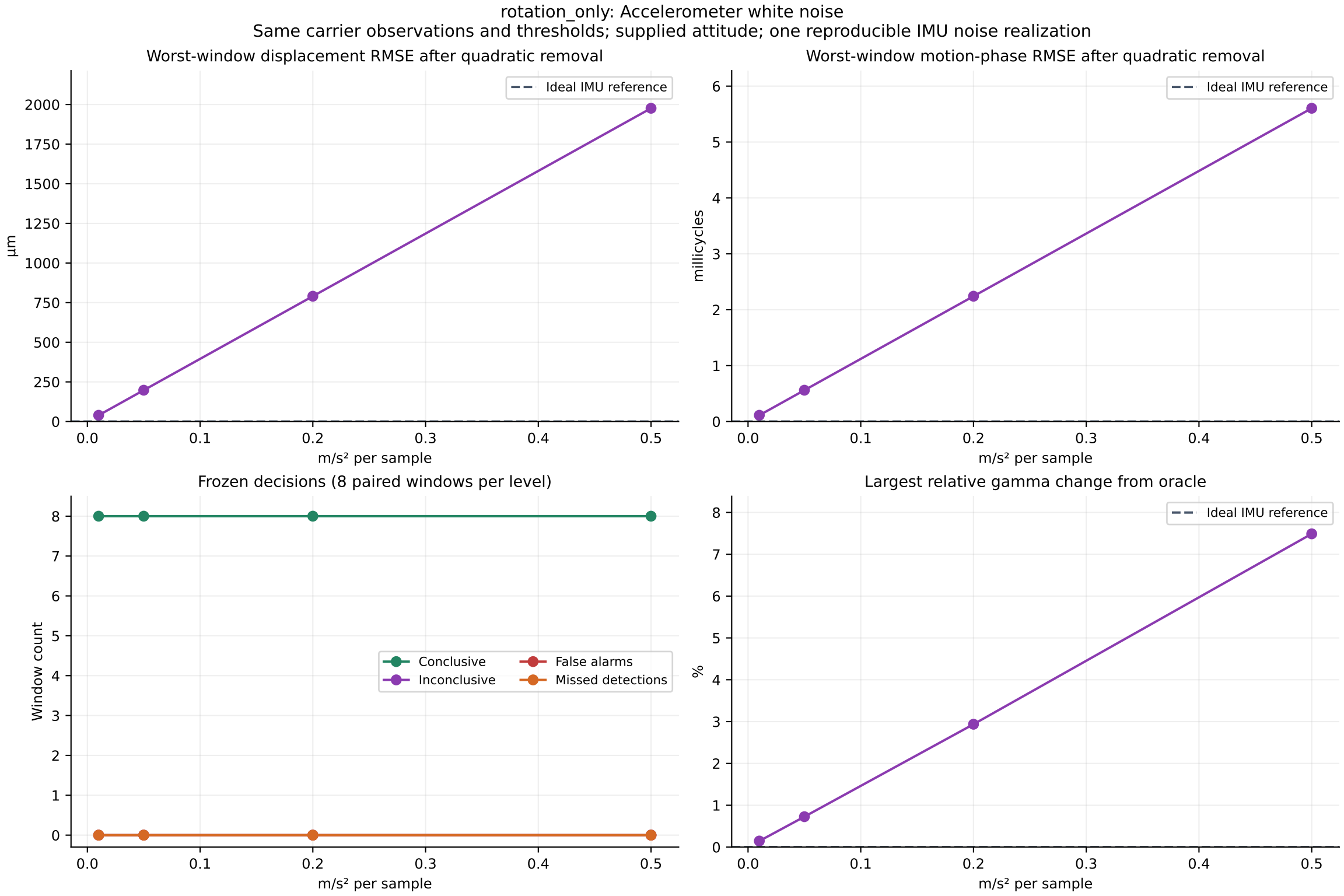

circle_coning: IMU timestamp offset
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
ms
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
ms
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
ms
Largest relative gamma change from oracle
1e−9
%
No comparable conclusive fits
ms

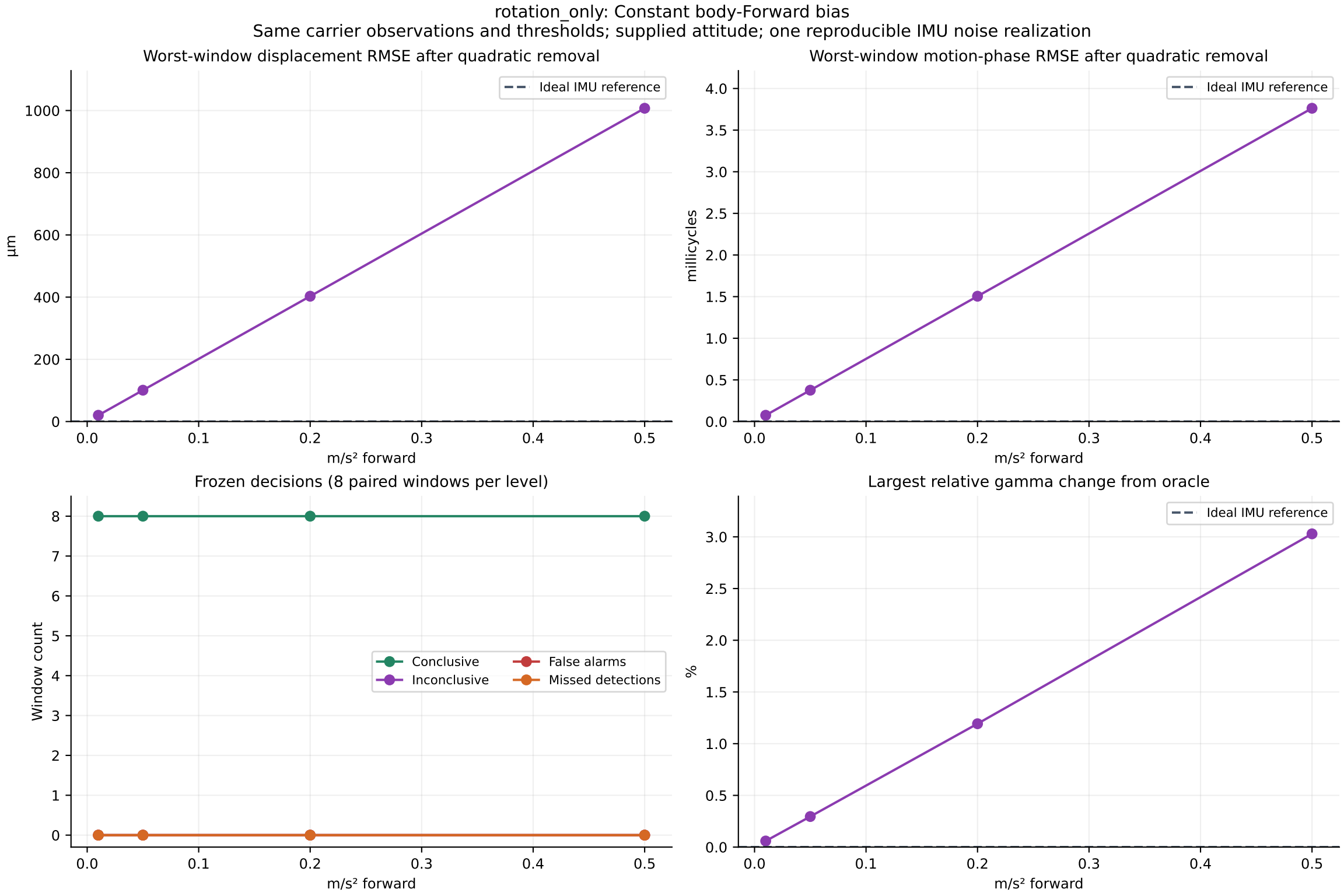

three_axis: Accelerometer white noise
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² per sample
Worst-window motion-phase RMSE after quadratic removal
Ideal IMU reference
millicycles
m/s² per sample
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
m/s² per sample
Largest relative gamma change from oracle
Ideal IMU reference
%
m/s² per sample

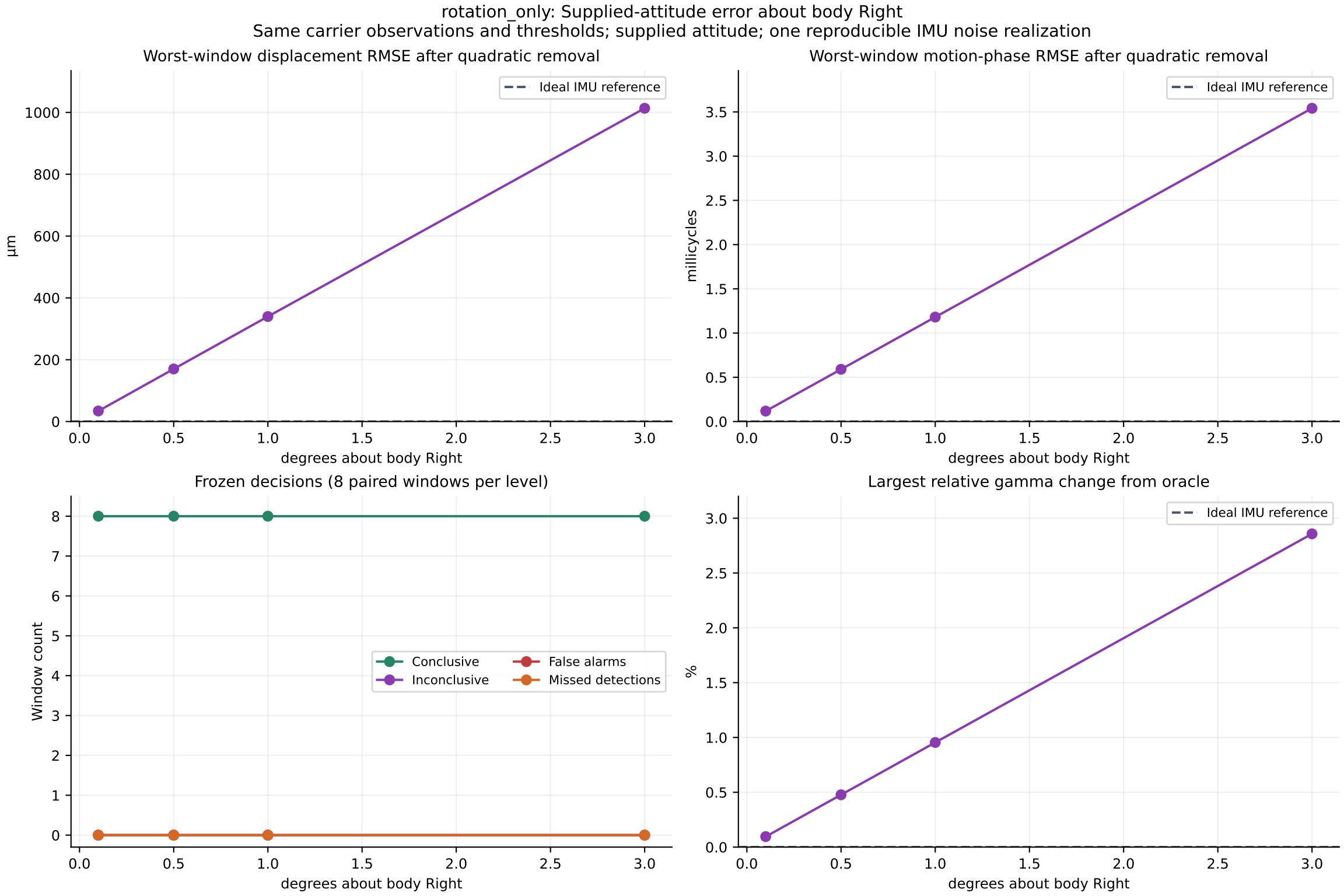

three_axis: Constant body-Forward bias
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
µm
m/s² forward
Ideal IMU reference
Worst-window motion-phase RMSE after quadratic removal
millicycles
m/s² forward
Ideal IMU reference
Frozen decisions (8 paired windows per level)
Window count
m/s² forward
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%
m/s² forward
Ideal IMU reference

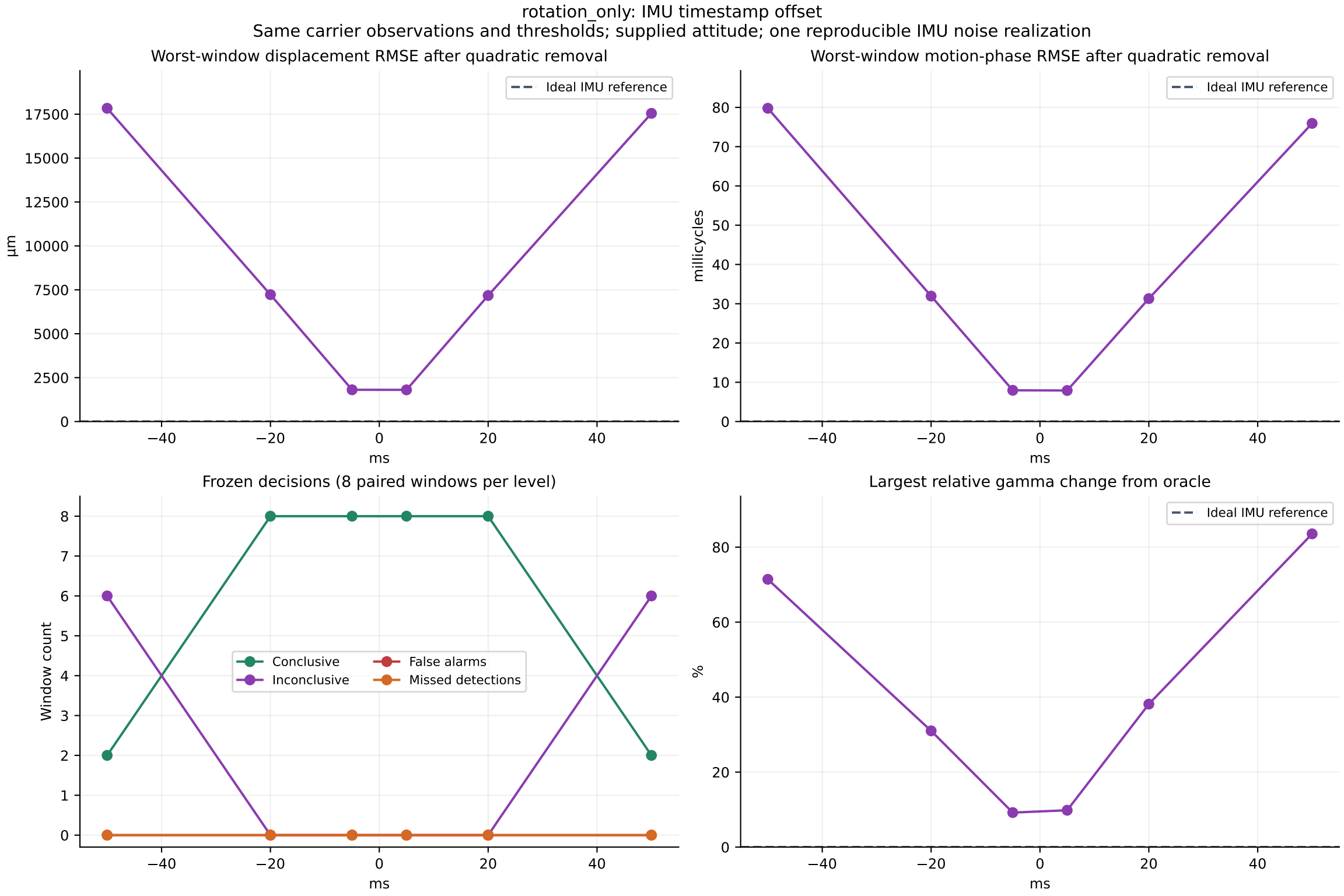

three_axis: Supplied-attitude error about body Right
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
µm
Ideal IMU reference
degrees about body Right
Worst-window motion-phase RMSE after quadratic removal
millicycles
Ideal IMU reference
degrees about body Right
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
degrees about body Right
Largest relative gamma change from oracle
%
Ideal IMU reference
degrees about body Right

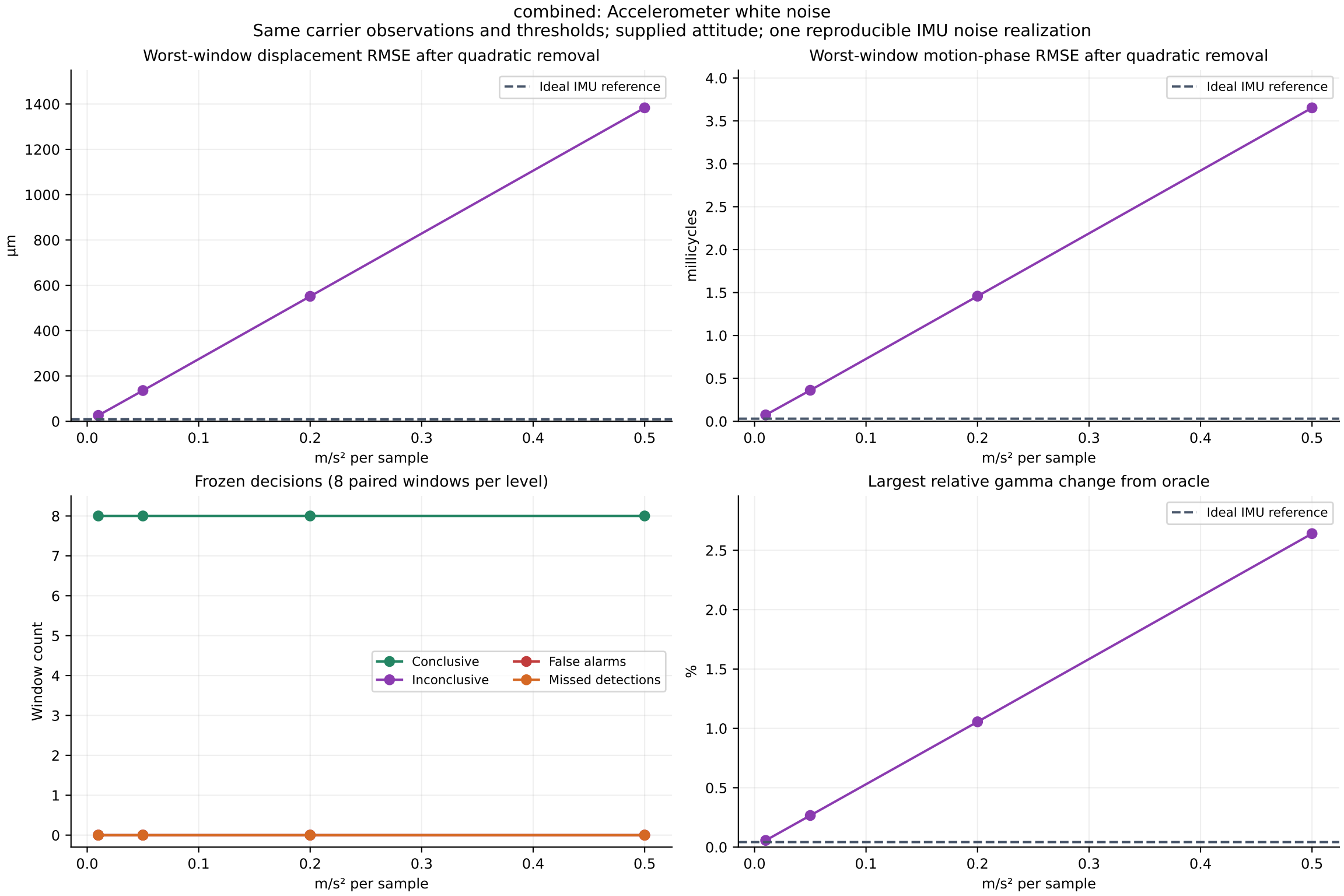

three_axis: IMU timestamp offset
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
µm
ms
Ideal IMU reference
Worst-window motion-phase RMSE after quadratic removal
millicycles
ms
Ideal IMU reference
Frozen decisions (8 paired windows per level)
Window count
ms
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%
ms
Ideal IMU reference

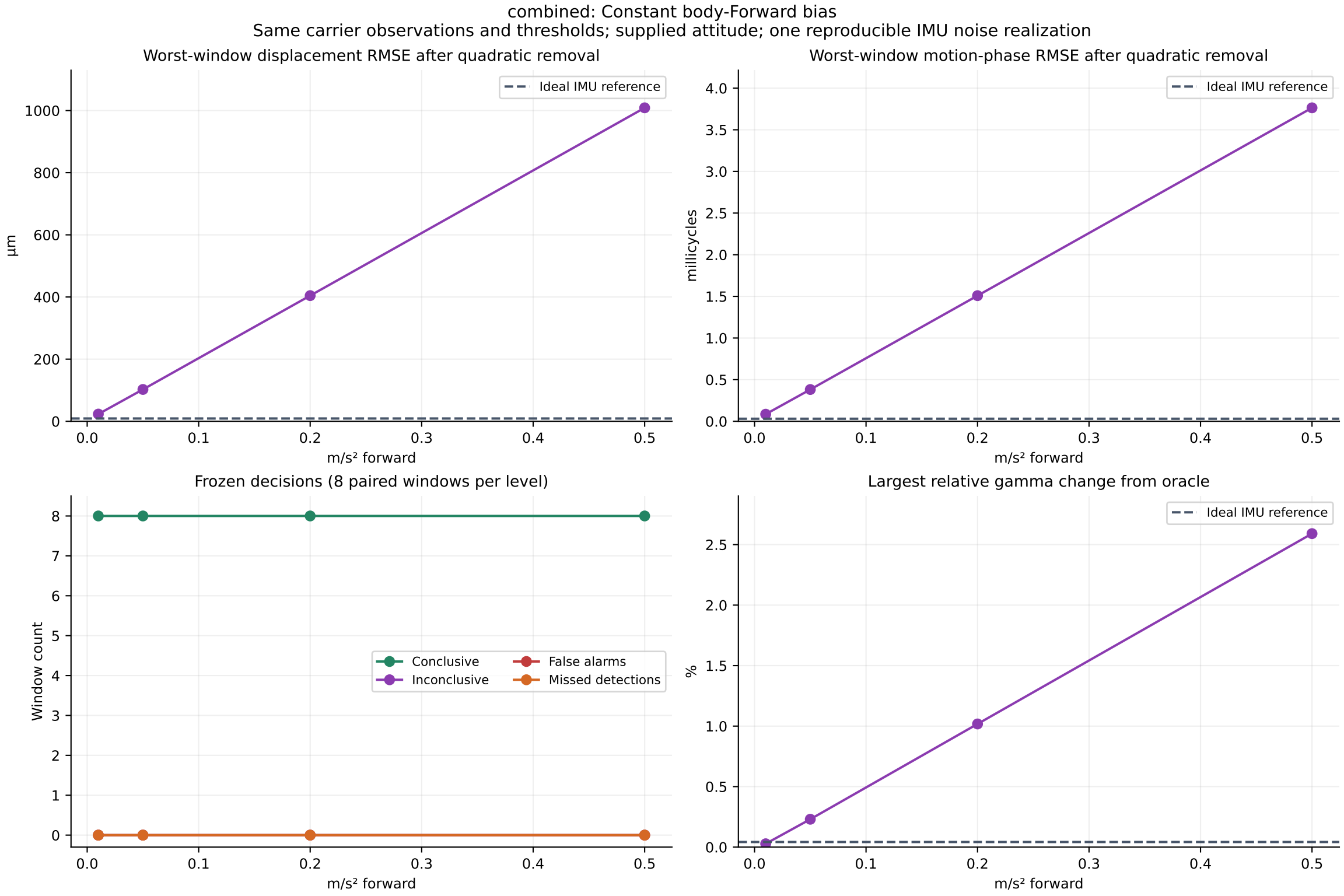

rotation_only: Accelerometer white noise
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
µm
m/s² per sample
Ideal IMU reference
Worst-window motion-phase RMSE after quadratic removal
millicycles
m/s² per sample
Ideal IMU reference
Frozen decisions (8 paired windows per level)
Window count
m/s² per sample
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%
m/s² per sample
Ideal IMU reference

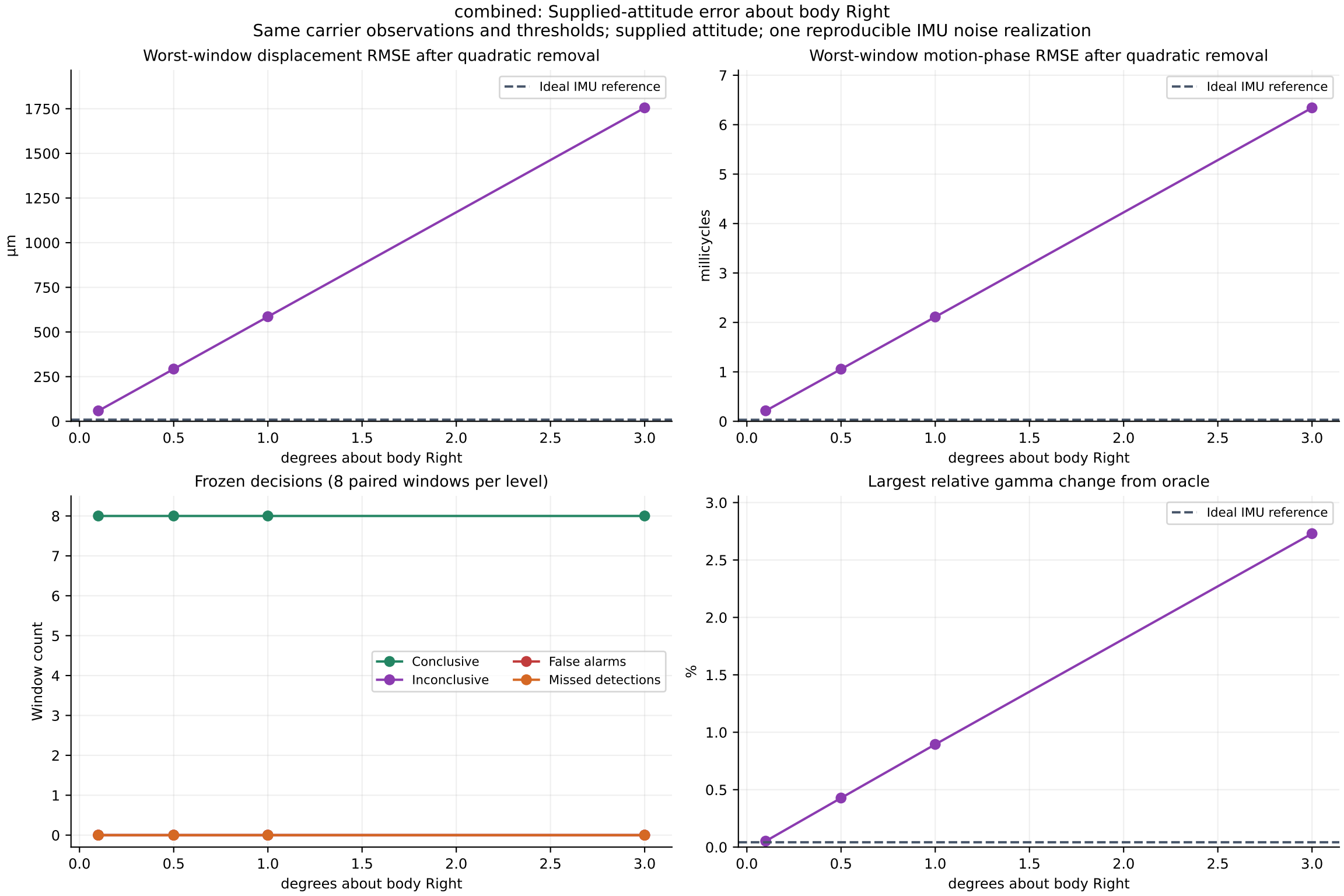

rotation_only: Constant body-Forward bias
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
µm
m/s² forward
Ideal IMU reference
Worst-window motion-phase RMSE after quadratic removal
millicycles
m/s² forward
Ideal IMU reference
Frozen decisions (8 paired windows per level)
Window count
m/s² forward
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%
m/s² forward
Ideal IMU reference

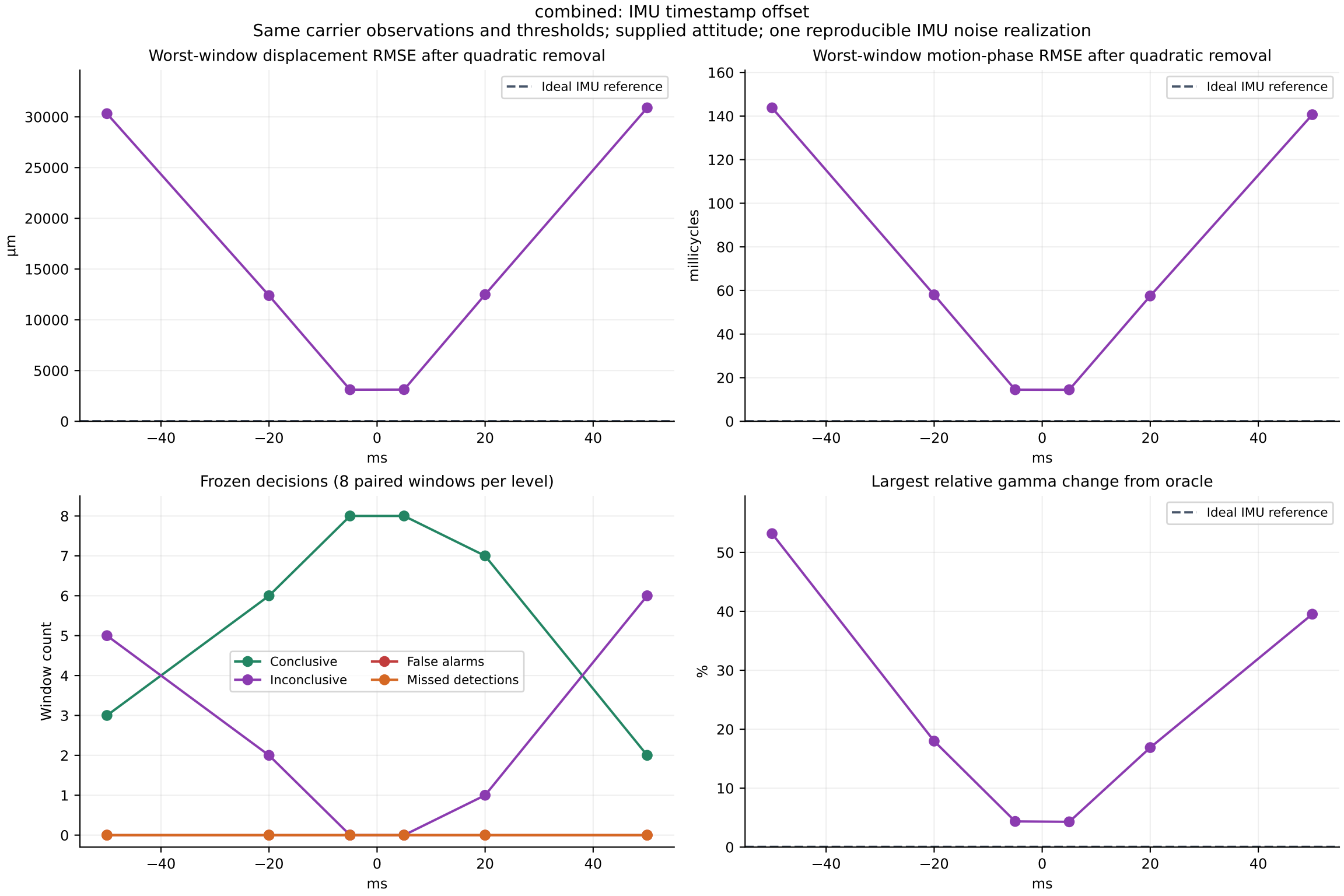

rotation_only: Supplied-attitude error about body Right
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
degrees about body Right
Worst-window motion-phase RMSE after quadratic removal
millicycles
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%


rotation_only: IMU timestamp offset
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
ms
Worst-window motion-phase RMSE after quadratic removal
millicycles
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%


combined: Accelerometer white noise
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² per sample
Worst-window motion-phase RMSE after quadratic removal
millicycles
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%


combined: Constant body-Forward bias
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
m/s² forward
Worst-window motion-phase RMSE after quadratic removal
millicycles
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%


combined: Supplied-attitude error about body Right
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
degrees about body Right
Worst-window motion-phase RMSE after quadratic removal
millicycles
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%


combined: IMU timestamp offset
Same carrier observations and thresholds; supplied attitude; one reproducible IMU noise realization
Worst-window displacement RMSE after quadratic removal
Ideal IMU reference
µm
ms
Worst-window motion-phase RMSE after quadratic removal
millicycles
Frozen decisions (8 paired windows per level)
Window count
Conclusive
Inconclusive
False alarms
Missed detections
Largest relative gamma change from oracle
%